

HUMAN RHINOVIRUS LEVERAGES MULTI-LAYERED IMMUNE EVASION MECHANISMS TO BYPASS HOST DEFENSES

Vyas Koduvayur¹, Priya Ravi², Rongying Zhang¹, Olivia Wu¹

¹*Department of Computational and Systems Biology, University of California, Los Angeles*

²*Department of Bioengineering, University of California, Los Angeles*

Keywords: Immune Evasion, Innate Immunity, Host-Virus Interaction, Viral Evolution, Antiviral Response, CpG Suppression, Codon Usage Bias, Host Transcriptional Response, Antigenic Variation, Sequence Divergence

ABSTRACT

Human rhinovirus (HRV), the leading cause of the common cold, remains a persistent global pathogen despite widespread exposure and repeated immune priming. With over 160 recognized types, HRV evades lasting immune memory and continues to cause recurrent infections throughout life. Despite extensive research, no vaccine has been developed due to its robust and multi-faceted immune evasion mechanisms. While HRV's prevalence is well recognized, the molecular strategies enabling its immune evasion remain poorly defined. In this study, we examine HRV's immune evasion strategies through four key layers: (1) genome composition, focusing on CpG suppression; (2) codon usage bias and its implications for immune recognition; (3) host transcriptional responses to HRV infection; and (4) evolutionary dynamics and sequence variability. While each of these aspects has been studied in isolation, their combined effects within HRV have not been fully explored. We also compare these findings with those from West Nile Virus, a zoonotic flavivirus that is a dead-end host in humans. Our results reveal a set of interlocking immune evasion strategies, including region-specific patterns of CpG suppression and TpG enrichment, codon bias driven by immune avoidance rather than translational efficiency, muted innate immune activation, and heightened sequence divergence likely shaped by sustained human selective pressures. These mechanisms form a complementary and interconnected framework of immune evasion tactics that reinforce one another, creating a cohesive strategy for HRV to evade immune detection and maintain persistence. Our findings support a paradigm shift toward mechanism-based frameworks in virology, guiding vaccine design and antiviral strategies beyond pathogen-specific models.

INTRODUCTION

Human rhinovirus (HRV) is a highly prevalent, positive-sense single-stranded RNA virus and is the primary causative agent of the common cold [1]. While HRV infections in healthy individuals are typically mild—manifesting as nasal congestion, sore throat, cough, headache, and low-grade fever—the virus can cause severe outcomes in vulnerable populations [2]. HRV is a leading cause of asthma exacerbations and chronic obstructive pulmonary disease (COPD) flare-ups, and it contributes significantly to lower respiratory tract infections in infants, the elderly, and immunocompromised individuals [3].

Despite being one of the most common viral pathogens globally, there is no approved vaccine for HRV. This is largely due to its extensive genetic diversity, antigenic variability, and the inability to establish a strong, long-term immune memory following infection [4]. HRV is classified within the *Picornaviridae* family and is divided into three major species—HRV-A, HRV-B, and HRV-C—each containing a genetically diverse array of circulating strains [2]. With over 160 recognized HRV types and little cross-protective immunity, individuals remain susceptible to repeated infections throughout their lives [5]. These challenges—combined with the economic costs HRV poses on global healthcare systems—underscore a critical need for deeper mechanistic insights into how this virus evades immune detection and clearance.

Although HRV has been studied for decades, the molecular mechanisms that allow it to persist despite frequent host exposure are not well understood. Unlike acute viral infections such as influenza or measles, HRV infections do not typically grant long-lasting immunity, and reinfections with the same or closely related strains are common [5]. This phenomenon suggests that HRV employs multiple, effective strategies to evade both innate and adaptive immune responses, but the extent to which these strategies interact is not well characterized. Understanding these tactics is crucial for informing the design of broad-spectrum antivirals, improving vaccine development, and deepening our understanding of host-pathogen coevolution.

Viral immune evasion can occur at multiple levels, including modulation of genome composition, host adaptation, and rapid evolution. Prior work has shown that many RNA viruses, including HRV, suppress CpG dinucleotides to avoid detection by certain host proteins and also exhibit codon usage biases that may enhance translation efficiency in human cells [6,7]. Other work has highlighted viral strategies that dampen host interferon responses and the role of rapid mutation in evading adaptive immunity [8,9]. While each of these mechanisms has been studied in isolation in other viruses, few studies have examined them side by side in HRV or systematically evaluated how they may operate in tandem. In this work, we explore HRV's immune evasion strategies through four complementary frameworks: (1) viral genome nucleotide composition, including CpG suppression; (2) codon usage bias and its implications for host translational efficiency; (3) transcriptional responses of the host immune system to HRV infection; and (4) evolutionary dynamics and genomic variability across HRV strains. To

contextualize our findings, we include a comparative analysis with *Orthoflavivirus nilense* (West Nile Virus), another positive-sense single-stranded RNA virus from the *Flaviviridae* family that is a dead-end host in humans and displays different patterns of immune engagement [10].

By integrating sequence-based, functional, and evolutionary analyses, we aim to better characterize the molecular and evolutionary strategies HRV uses to persist in human hosts. Our findings can help identify individual molecular signatures of immune avoidance, reveal how different evasion tactics may reinforce one another, and evaluate whether HRV's success is driven by a single dominant strategy or a coordinated set of immune evasion mechanisms. In doing so, we can provide new insights that could inform future vaccine development, contribute to the design of broad-spectrum antiviral therapies, and improve our understanding of how persistent viruses successfully navigate host immunity.

METHODS

We selected 16 HRV-C strains and 12 West Nile virus (WNV) strains (complete genome) spanning the past 25 years to analyze. WNV, a positive-sense single-stranded RNA virus like HRV, was chosen as a comparison group due to its distinct transmission pattern—it primarily circulates in bird populations and is transmitted by mosquitoes, with humans serving as dead-end hosts [10]. FASTA files of viral nucleotide sequences, stored as complementary DNA (cDNA), were sourced from the National Institutes of Health (NIH) GenBank database [11]. Accordingly, all thymine (T) bases in the sequences reflect uracil (U) in the original RNA genomes. Throughout this paper, references to "T" correspond to the cDNA representations found in GenBank. Additional similar strains to the initial reference were identified through BLAST searches [12]. Accession numbers can be found in the data availability section.

Genomic Signature Analysis

The Biostrings package was used to process and clean FASTA sequences [13]. Average frequencies of all dinucleotide, trinucleotide, and tetranucleotide sequences were found for both viruses and used to calculate the Observed/Expected (O/E) ratio across the entire genome:

$$O/E \text{ Ratio} = \frac{f_{obs}(k)}{\prod_{i=1}^k f(n_i)}$$

where $f_{obs}(k)$ is the observed frequency of the k-mer, $f(n_i)$ is the frequency of the individual nucleotide at position i, and $\prod f(n_i)$ is the expected frequency assuming independent nucleotide distribution.

This metric was chosen as a standardized method to assess whether a sequence is underrepresented or overrepresented relative to what would be expected based on the random

distribution of nucleotides. AT and GC content (the proportion of adenine/thymine and guanine/cytosine nucleotides, respectively) were also calculated and averaged across all strains for both viruses.

Further analysis of O/E ratios across HRV strains was performed for two particularly notable cases: (1) the CpG dinucleotide, which was found to be especially underrepresented in HRV strains and (2) the TpG dinucleotide, which was found to be especially overrepresented. Average O/E ratios for each motif then were calculated across specific genomic features, with the protein-coding regions defined by the start and end nucleotide sequences as annotated by GenBank. Viral capsid proteins (VP1, VP2, VP3, and VP4) were each analyzed separately, and all other non-structural proteins (2A, 2B, 2C, 3A, 3B, 3C, 3D) were grouped together.

Codon Adaptation Analysis

We compared the frequency of each codon in the HRV genome to that of the human genome to identify biases in virus translation machineries and provide insight into viral adaptation and replication fitness. Coding regions from the FASTA sequences of the same HRV and WNV strains used above were processed with Biostrings. Coding regions were specifically selected due to their influence on protein synthesis and relevance for examining translational efficiency. Sequences were split into codons, and their relative frequencies were computed. These frequencies were compared to a human codon frequency table to assess differences in usage frequencies.

To evaluate how efficiently HRV genes are translated within the human host, we used the Codon Adaptation Index (CAI), a metric that estimates gene expression potential based on codon usage bias [14]. CAI quantifies how closely a genome's codon usage matches a reference codon usage table—in this case, a general human codon frequency table from the publicly available GenScript tool (<https://www.genscript.com/tools/codon-frequency-table>).

Each of the 64 codons is assigned a “relative adaptiveness” value, w , which reflects its frequency relative to the most frequent synonymous codon for the same amino acid in the reference set. These relative adaptiveness values range from 0 to 1, with a value of 1 indicating that the codon occurs most often for a given amino acid and a value of 0 indicating complete absence. The relative adaptiveness value is found with

$$w_{aa,i} = \frac{f_{aa,i}}{f_{aa,max}}$$

where $w_{aa,i}$ represents the relative adaptiveness value of the i_{th} codon, $f_{aa,i}$ represents the frequency of that codon in the HRV genome, and $f_{aa,max}$ represents the frequency of the most frequent synonymous codon in the host genome.

The CAI of a virus strain is then calculated as the geometric mean of the relative adaptiveness values across all codons in the gene:

$$CAI = \prod_{i=1}^N w_i^{1/N}$$

where w_i represents the relative adaptiveness of the i_{th} codon and N represents the total number of codons present in both genes. This value serves as a predictor of how efficiently the gene might be translated in the host genome, with higher CAI scores generally correlating with higher adaptation levels. CAI values were calculated for each HRV and WNV strain.

RNAseq and Microarray Data Access

To investigate host transcriptional responses to viral infection, we analyzed publicly available gene expression datasets from the Gene Expression Omnibus (GEO) database [15], focusing on human macrophages due to their central role in initiating and regulating innate immune responses [16]. We separately analyzed both HRV and WNV infection models.

The WNV dataset (GSE51808) contained RNA sequencing (RNAseq) data profiling gene expression in primary human macrophages from 10 healthy donors infected *in vitro*. We selected the first 10 samples—5 infected with WNV and 5 controls—to identify differentially expressed genes (DEGs).

Since no suitable RNAseq data was available for macrophage responses to HRV, we instead analyzed a microarray-based dataset (GSE106963) that profiled gene expression using the Affymetrix Human Gene 2.1 ST Array [17]. The original study examined macrophages under three polarization states—unpolarized, M1-polarized, and M2-polarized—each exposed to either HRV-A16 or a sham control. We selected a subset of 10 unpolarized macrophage samples—5 infected with HRV-A16 and 5 controls—in order to restrict analysis on virus-induced transcriptional changes and reduce confounding effects of polarization.

Differential Gene Expression Analysis

For WNV, low-count genes (≤ 1 read across all samples) were filtered out before normalization. DEGs were found using the DESeq2 package [18], and significant genes were identified based on an absolute $\log_2$ fold change ($\log_2FC$) > 1 and adjusted p-value < 0.05 . For HRV, probe-level intensities from the microarray dataset were processed using RMA normalization with the oligo package [19], and DEGs were found with limma [20]. Significant genes were identified as described above. Functions of select top differential genes were sourced from the GeneCards database [21].

Pathway Enrichment Analysis

DEG results from previous steps were used to construct ranked gene lists ordered by log2FC. To identify biological processes enriched in response to viral infection, Gene Set Enrichment Analysis (GSEA) was performed via the `gseGO()` function from the `clusterProfiler` package (parameters: `ont="BP"`, `minGSSize=10`, `maxGSSize=100`, `pAdjustMethod="BH"`) [22]. Normalized Enrichment Scores (NES) and adjusted p-values were then extracted and filtered to highlight specific immune-related Gene Ontology terms of interest. GSEA was chosen over other overrepresentation methods to account for the full spectrum of gene expression changes and to emphasize the magnitude of log2FCs due to differing numbers of DEGs across groups.

Phylogenetic Tree and Clustering Analysis

A separate set of 54 HRV sequences, distinct from the previously described 16 strains, was analyzed. These sequences were collected in the United States between 2016 and 2024 and were selected based on the availability of complete genome data and metadata. Similarly, 79 WNV sequences with valid US collection dates between 1999 and 2008 were included. Although the year ranges differ, both datasets span comparable 8–9 year periods selected to capture strain diversity based on the publicly available complete genome data. In both cases, a reference genome was retrieved from NCBI GenBank [11], and similar sequences were identified using the BLAST tool [12]. Accession numbers for these strains can be found in the Data Availability section.

Metadata for all sequences were curated in tab-separated format using the Entrez Programming Utilities (E-utilities) API. Multiple sequence alignments were performed using the DECIPHER package in R, which is optimized for aligning highly divergent sequences [23]. Alignment parameters were set to 100 iterations and 50 refinement steps to ensure accurate resolution of insertions, deletions, and single nucleotide polymorphisms. The final alignments were exported in FASTA format for downstream analyses.

To infer evolutionary relationships among the 54 HRV strains, a Maximum Likelihood (ML) phylogenetic tree was constructed using the `phangorn` package in R [24]. Sequences were converted to `phyDat` format, and an initial tree was generated using the Neighbor Joining method. The ML tree was then fitted using the General Time Reversible (GTR) nucleotide substitution model and optimized with stochastic rearrangement to explore alternative topologies. This approach allowed for the joint estimation of tree topology and branch lengths under a statistically rigorous framework, modeling sequence evolution under realistic mutation dynamics. Final tree branches were colored by year of collection to highlight temporal structure.

To further investigate genetic relationships and potential temporal patterns, we applied Uniform Manifold Approximation and Projection (UMAP) using the `uwot` package in R [25]. A pairwise nucleotide distance matrix was computed from the aligned sequences using Hamming distance,

which quantifies the proportion of differing nucleotide sites between each sequence pair. This metric, which treats nucleotide bases as categorical variables, is well-suited for capturing raw dissimilarities without assuming a specific evolutionary model. The resulting distance matrix was used as input for UMAP ($n_neighbors = 10$ and $min_dist = 0.5$). UMAP projection was performed using Euclidean distance.

Sequence Divergence Analysis

To estimate the rate of sequence divergence for HRV and WNV, a reference-based linear modeling approach was applied. For each virus, the earliest strain based on collection date was selected as the reference sequence. Mutation counts were computed by comparing each sequence to the reference, counting nucleotide differences at aligned positions while excluding gaps. This approach provided a straightforward measure of sequence divergence (i.e., mutations) relative to the reference strain over time. Mutation counts were then merged with collection metadata to calculate the number of years elapsed since the reference strain for each sample. A linear regression model was then fitted using the formula:

$$Mutations = \beta_0 + \beta_1 \times years\ since$$

where β_0 represents the baseline number of mutations (set to zero for the reference strain) and β_1 represents the average number of mutations that accumulate per year since the reference strain. To obtain a normalized rate of divergence (mutations per year), the slope β_1 was divided by the length of the reference genome to yield a per-base, per-year divergence rate.

Statistical Analysis

All analysis was performed with R (v. 4.4.1). Error bars on all figures represent +/- 1 standard deviation from the mean. Non-parametric Wilcoxon rank-sum tests were used to assess statistical significance between groups, and Benjamini-Hochberg multiple correction testing was used when appropriate. Significance was determined with a p-value <0.05.

RESULTS

CpG Suppression and TpG Enrichment Drive Distinct Nucleotide Biases in Rhinovirus

Average O/E ratios for each dinucleotide and trinucleotide motif across all HRV strains were visualized (**Figure 1A**). Across dinucleotides, the CpG dinucleotide motif was the most underrepresented, with an average O/E ratio of 0.30799. A Wilcoxon rank-sum test affirmed a significant difference between the O/E ratio of the CpG motif and all other dinucleotides ($p=1.663671e-09$). Across trinucleotides, the 8 CpG-containing motifs were found to have the

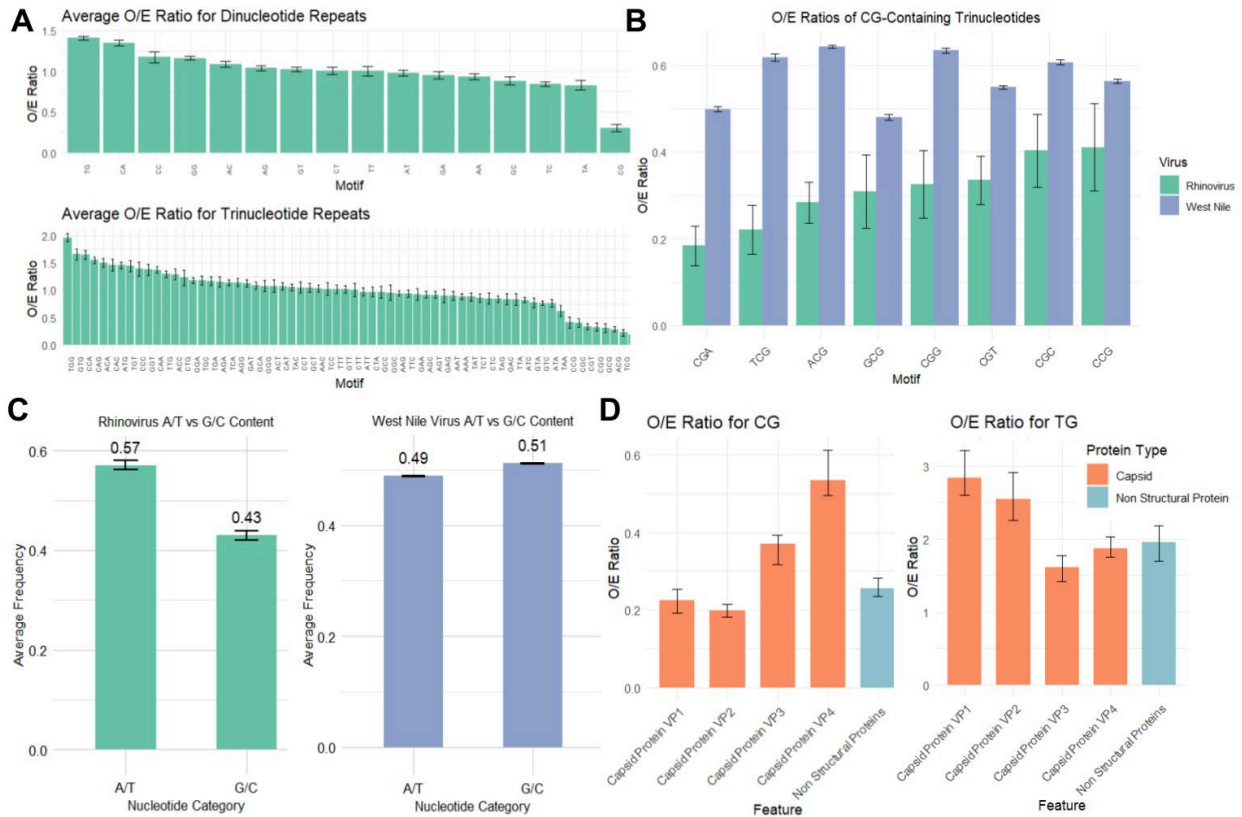


FIGURE 1: Rhinovirus Genome is Defined by Region-Specific Underrepresentation of CpG and Enrichment of TpG.

- Bar chart (with error bars) of average observed/expected (O/E ratio) across 16 Rhinovirus strains for each dinucleotide and trinucleotide motif in its genome.
- Bar chart with error bars of average O/E ratio of each CpG-containing trinucleotide motif for Rhinovirus (16 strains) and West Nile Virus (12 strains).
- Bar charts (with error bars) of average AT and GC content in Rhinovirus and West Nile genomes across strains.
- Bar chart (with error bars) of average O/E ratio for CpG and TpG dinucleotides for each feature-coding region across Rhinovirus strains.

lowest O/E ratios, with the Wilcoxon rank-sum test finding a significant difference between this group and all other trinucleotides ($p=2.852485e-06$). Additionally, the TpG dinucleotide and the TGG trinucleotide motifs (noting that T represents U in the original RNA genome) were found to be the most enriched in the HRV strains, with average O/E ratios of 1.4133 and 2.0668, respectively. Similar patterns were observed when looking at tetranucleotide repeats as well (including CpG suppression and TpG enrichment).

The O/E ratios of the eight CpG-containing trinucleotides for HRV were then plotted against those of WNV (**Figure 1B**). The ratios for all eight trinucleotides were much lower in HRV compared to WNV, and a significant difference between the two groups was observed ($p=9.04e-06$). Additionally, AT and GC content was plotted for both viruses (**Figure 1C**). The average AT:GC ratio for HRV was found to be 57:43, which is comparable to that of humans (~60:40) [26]. In contrast, the ratio for WNV was significantly different ($p=8.33e-07$) with a ratio around 49:51.

Finally, the O/E ratios for the CpG dinucleotide and the TpG dinucleotide were also found across several protein-coding features (VP1, VP2, VP3, VP4, and non structural proteins) in HRV (**Figure 1D**). The CpG motif was low throughout all regions but was particularly underrepresented in regions coding for VP1, VP2, and the non structural proteins, and less so in regions coding for VP4. Conversely, the TpG motif was overrepresented throughout the entire genome but was especially enriched in VP1 and VP2, marking a notable parallel.

Rhinovirus Exhibits Lower Codon Adaptation to Humans Compared to West Nile Virus

The codon GAT (with T representing U in the original RNA genome), which codes for aspartic acid, was the most frequent in the HRV genome with a frequency of 0.0357. TAG, a stop codon, was the least frequent with a frequency of $8.76e-5$ (**Figure 2A**). Compared to the human genome, the codons ACA and ATA (threonine and isoleucine, respectively) were found to be the most overused in the HRV genome, while CTG and GAG (leucine and glutamate) were the most underused. The codons CTT, GCT, and CAT (leucine, alanine, and histidine) were the closest to equally represented in both genomes (**Figure 2B**). In the WNV genome, the GGA codon (glycine) was the most overused, and the codon CAG (glutamine) was the most underused. TAG and CAA (stop and glutamine) were nearly equally represented (**Figure 2C**).

The mean CAI was found to be 0.588 ± 0.014 across the 16 HRV strains and 0.626 ± 0.00070 across the 12 WNV strains (**Figure 2D**), indicating that WNV exhibits higher adaptation efficiency with this metric. A Wilcoxon rank-sum test determined there to be a statistically significant difference between the two groups ($p=9.247e-06$).

West Nile Virus Elicits Broader Innate Immune System Signature Than Rhinovirus

Significantly upregulated and downregulated genes from microarray and RNA-seq datasets were analyzed in macrophages infected with HRV-A16 and WNV, respectively (**Figure 3A**). HRV infection led to the induction of several classic type I interferon-stimulated genes (IFIT1, RSAD2, IFI44L) as well as chemotactic genes (CXCL10, GBP1), suggesting recruitment of immune cells to sites of infection. WNV infection triggered direct expression of Type I interferons themselves (IFNB1, IFNA2, IFNW1) in addition to chemokines (CXCL10, CXCL11), indicating a broader activation of upstream antiviral signaling pathways. Notably repressed genes varied in function but generally clustered around pathways related to cell death

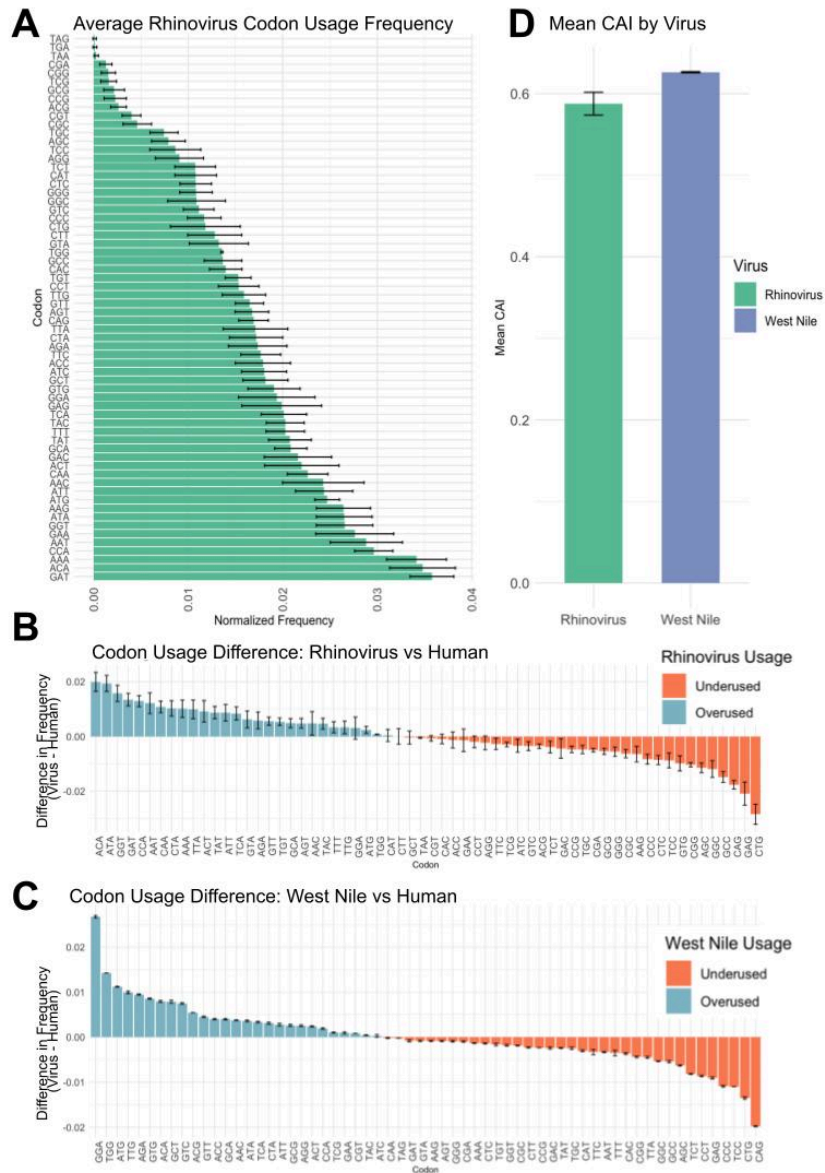


FIGURE 2: *West Nile Virus Exhibits Greater Codon Adaptation Than Rhinovirus Despite Human Host Specialization.*

- Bar chart (with error bars) of the average frequency of each codon across 16 Rhinovirus strains.
- Mean difference chart (with error bars) of the difference in codon usage frequency between the Rhinovirus and human genomes (Rhinovirus - human).
- Mean difference chart (with error bars) of the difference in codon usage frequency between the West Nile Virus and human genomes (West Nile - human).
- Bar chart (with error bars) of average CAI values across the 16 Rhinovirus strains and 12 West Nile strains.

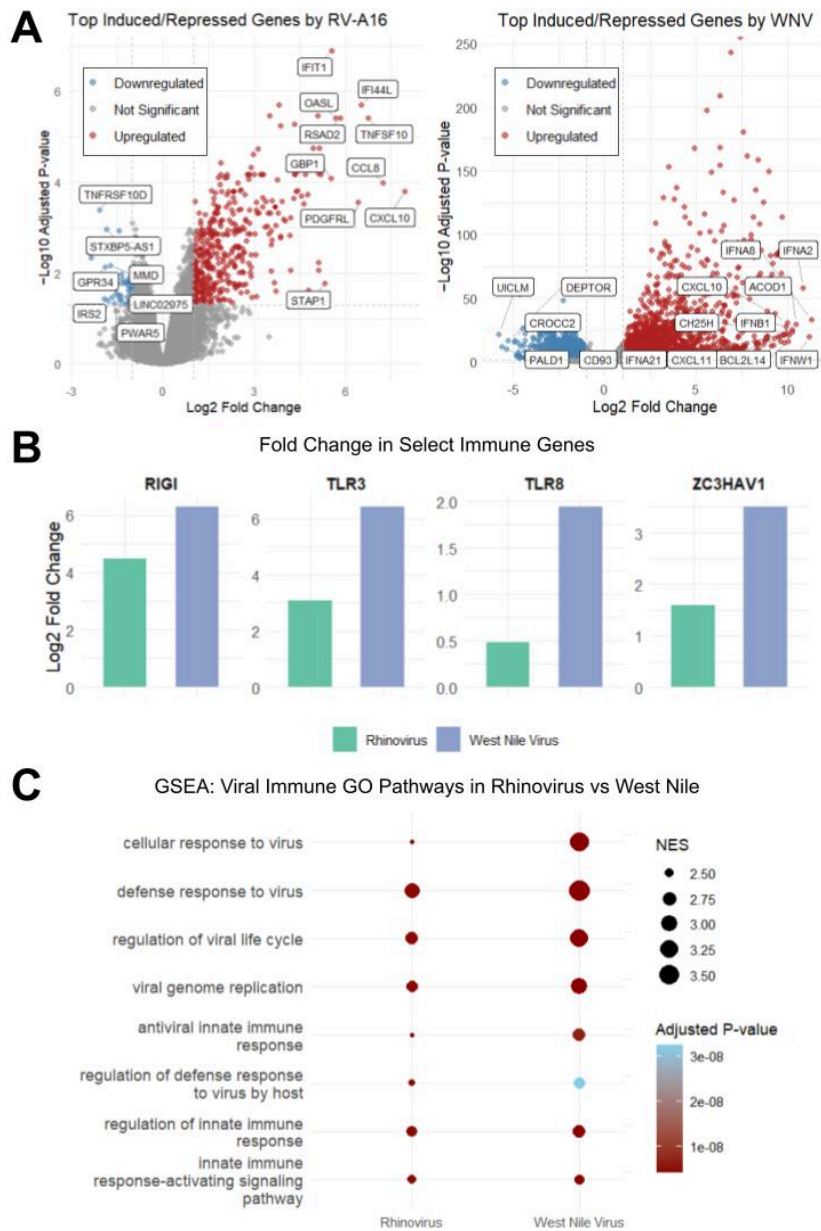


FIGURE 3: *West Nile Virus Induces More Extensive Activation of Antiviral and Viral Sensing Pathways Than Rhinovirus*

- Volcano plots of significant differentially expressed genes ($\log_2\text{FC} > 1$, adjusted p-value < 0.05) following Rhinovirus-A16 and West Nile Virus infection in human macrophages.
- Bar graphs of Rhinovirus and West Nile Virus $\log_2\text{FC}$ s for genes associated with single-stranded RNA viruses sensing: RIGI, TLR3, TLR8, and ZC3HAV1.
- Gene Set Enrichment Analysis for select viral and immune-related Gene Ontology biological pathway terms showing (1) normalized enrichment scores (NES) and (2) adjusted p-values.

and survival (TNFRSF10D, DEPTOR, IRS2) and immune modulation (GPR34). A full summary of the functions of genes displayed in the volcano plots can be found in tabular form (**Table 1**).

To assess differential activation of innate viral sensing pathways, log2FCs were compared for 4 specific genes associated with sensing of single-stranded RNA viruses: RIG-I, TLR3, TLR8, and ZC3HAV1 (**Figure 3B**). All four genes were more strongly induced by WNV compared to HRV, with TLR8 showing an especially pronounced difference—being significantly upregulated in WNV infection but not significantly changed in HRV.

Normalized enrichment scores (NES) and adjusted p-values were calculated and plotted for eight biological process pathways related to viral immune sensing and innate immunity (**Figure 3C**). An NES greater than 1 indicates positive enrichment of a gene set within the ranked expression data. While both viruses activated all of these pathways, WNV exhibited stronger enrichment overall, with notably higher NES scores for “cellular response to virus” and “defense response to virus.” A full list of all DEGs and NES scores associated with each GO term can be found in tabular form (**Table 2**).

Rhinovirus Exhibits Increased Sequence Divergence Relative to West Nile Virus

Divergence rates were estimated by measuring nucleotide differences between each strain and the earliest (reference) strain in the dataset. HRV strains exhibited a divergence rate of 0.0122 nucleotide differences per base per year, while WNV strains showed a much lower divergence rate of 0.00011 nucleotide differences per base per year (**Figure 4A**).

To explore genetic relationships among HRV strains, we applied UMAP to the pairwise nucleotide distances. The resulting two-dimensional UMAP embedding revealed distinct groupings of strains based on their collection year, with more recent sequences positioned on the right side of the plot (**Figure 4B**).

A maximum likelihood phylogenetic tree provided additional insights into the evolutionary relationships of the HRV strains (**Figure 4C**). Early strains from 2016 and 2019 dispersed across the tree, while recent strains from 2023 and 2024 formed distinct clusters after the first major divergence, indicating the emergence of new lineages. Notably, some older strains remained on separate branches, often with newer strains appearing at subsequent splits. Branch lengths reflected varying sequence divergence, and strains from similar years clustered closely.

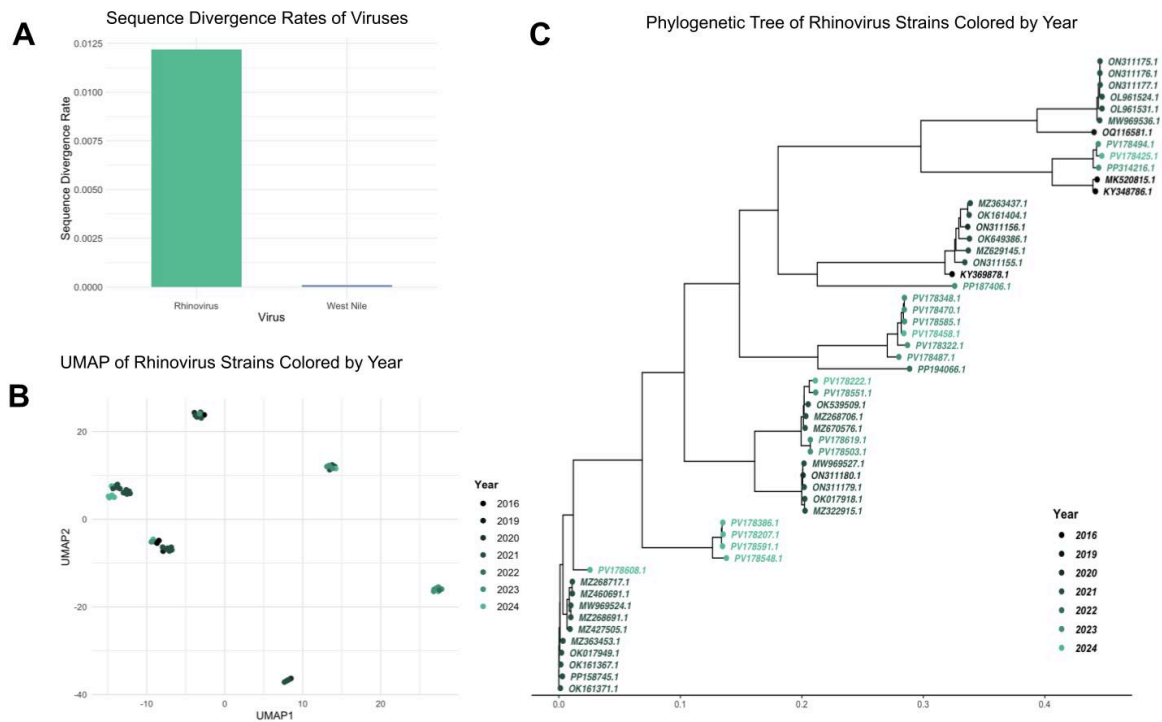


FIGURE 4: Rhinovirus Shows Greater Sequence Divergence Rates than West Nile Virus

- A. Bar plot comparing the estimated divergence rates (average number of mutations accumulated per base per year) across 54 Rhinovirus strains (2016–2024) and 79 West Nile Virus strains (1999–2008) collected in the US.
- B. UMAP projection of the 54 Rhinovirus strains colored by collection year.
- C. Maximum Likelihood phylogenetic tree of the 54 Rhinovirus strains using a Neighbor-Joining tree as base model. Accession numbers from NCBI are shown for each strain, and tips are colored by year.

DISCUSSION

CpG Suppression and TpG Enrichment in Rhinovirus Genomes Function as a Coordinated, Region-Specific Strategy for Immune Evasion

CpG suppression across the viral genome is a phenomenon that has been well documented in several pathogenic single-stranded RNA viruses like influenza and SARS-CoV-2 [27,28]. Our findings align with prior literature that have previously demonstrated that CpG suppression is an immune evasion strategy that likely evolved as an evolutionary response to the low CpG content in mammalian hosts due to methylation and deamination processes that convert CpG into a TpG or CpA dinucleotide [29]. The deamination process has led to less frequent CpG sites in the host genome, but it also has been shown to serve an immune function: certain pattern recognition

molecules recognize CpG motifs as foreign, which can trigger an immune response [30]. For example, a well-characterized mechanism involves the zinc-finger antiviral protein (ZAP), which specifically binds to CpG-rich RNA and targets it for degradation [31]. In response, RNA viruses like HRV can in turn evolve to also suppress these motifs in order to avoid immune detection, leading to low CpG content across their genomes (**Figure 1A**).

We also observed a notable enrichment of the TpG and CpA motifs throughout the HRV genome in both dinucleotides and trinucleotides (**Figure 1A**). This pattern aligns with the well-established theory that CpG motifs are thought to mutate into other dinucleotides like TpG or CpA [32]. Our results align with prior literature that has found CpG suppression and TpG enrichment to be associated, suggesting that as CpG content is reduced, TpG (along with CpA) becomes more abundant. Notably, CC and GG dinucleotides also show increased frequencies, suggesting that these are also mutation products of CpG motifs. This overall shift in dinucleotide composition highlights CpG suppression as the major driver influencing the differences in observed O/E ratios across the HRV genome.

The lower O/E ratios for CpG-containing trinucleotides in HRV compared to WNV suggest that HRV may be under especially strong selective pressure to suppress CpG motifs (**Figure 1B**). Unlike WNV, which is transmitted through mosquito vectors and undergoes replication in both avian and mammalian hosts, HRV is a strictly human pathogen. The observed CpG suppression may reflect increased adaptation to the human immune environment, while viruses with broader host ranges may not experience the same level of selective pressure. This distinction reinforces the idea that CpG suppression is not just a passive consequence of viral evolution but an active strategy to enhance viral fitness in a specific host environment.

Although CpG suppression has been extensively characterized for its role in immune evasion, it has not yet been quantitatively linked to viral replication efficiency in human populations. Our findings suggest that the level of CpG suppression can serve as a quantitative marker for immune evasion strength, with stronger suppression correlating with greater viral replication efficiency in human hosts. Since HRV replicates efficiently and spreads widely in human populations, its extreme CpG suppression may be a key evolutionary feature that enables sustained transmission. In contrast, WNV, which does not rely on human-to-human transmission, exhibits higher (even if somewhat suppressed) CpG levels, further supporting the idea that CpG suppression is a mark of viruses highly adapted to evading human immune defenses.

Further comparative analysis between HRV and WNV revealed that the AT:GC ratio in HRV closely resembles that of humans whereas WNV shows a significantly different composition (**Figure 1C**). This similarity between HRV and human nucleotide composition suggests another immune evasion strategy—mimicry of host genomic content. Viruses with nucleotide compositions that resemble their host's may evade innate immune sensors that detect deviations

in nucleotide frequencies, such as those involved in RNA editing and recognition of foreign RNA. This may contribute to the ability of HRV to establish itself and persist in the host despite constant immune surveillance.

CpG suppression was observed across all regions of the HRV genome but was particularly noticed in genomic sequences encoding VP1, VP2, and non-structural proteins (**Figure 1D**). This pattern suggests that different coding regions of the viral genome experience varying levels of selective pressure to reduce CpG content, likely due to the differing immunological exposure and functional importance of the proteins they encode. VP1 and VP2 are major surface-exposed capsid proteins that play essential roles in mediating viral attachment and entry into host cells, and efficient expression of these proteins is critical for infectivity and immune evasion [33]. If the RNA encoding these proteins is degraded through mechanisms like ZAP binding due to high CpG content, it could directly impair the virus's ability to infect cells or avoid immune detection, resulting in a major fitness loss. In contrast, VP4 is located on the interior of the capsid and is not accessible to the extracellular immune environment [33]. Although its expression is still necessary, the consequences of reduced VP4 levels may be less severe as it doesn't interact directly with host immune defenses, leading to a weaker selective pressure to suppress CpG in its coding sequence.

Similarly, the TpG motif was overrepresented everywhere within the open reading frame, but especially so in regions coding for VP1 and VP2 compared to VP4. The strong presence of TpG in the surface protein-coding regions could be seen as a direct result of CpG motifs mutating into TpG to evade host immune recognition. This pattern reinforces the notion that CpG suppression and TpG enrichment function as coordinated, region-specific mechanisms that are driven by selective pressures on the virus to optimize its immune evasion strategies. The positional constraints of the virus, particularly in surface-exposed proteins like VP1 and VP2, further influence these patterns, with the viral genome evolving to suppress CpG and therefore enrich TpG in regions coding for proteins that play more essential roles. Together, these patterns underscore how HRV has evolved to optimize its replication and persistence in the human host, with both CpG suppression and TpG enrichment contributing to its immune evasion and survival.

Immune Avoidance, Not Translation Efficiency, Drives Codon Bias in Rhinovirus

GAT, which codes for aspartic acid, was found to be the most frequent codon in the HRV genome (**Figure 2A**). Notably, GAT is only present moderately in the WNV. Aspartic acid has been shown to play a significant role in transporting electrons into the mitochondria, thus regulating glycolysis and maintaining the cell's redox balance [34]. A study on foot-and-mouth disease virus (FMDV), a member of the *Picornaviridae* family like HRV, demonstrated that aspartic acid levels increased during FMDV infection [35]. Supplementing cells with aspartate was found to enhance FMDV replication in a dose-dependent manner. Similar effects were

observed with enterovirus 71 (EV71) and Seneca Valley virus (SVV), indicating that aspartate supports viral replication across multiple picornaviruses [35]. In poliovirus, another picornavirus, aspartic acid residues are found to be crucial for viral RNA-dependent RNA polymerase (3Dpol) function. Mutations in conserved aspartic acid residues within the GDD motif of 3Dpol significantly alter the enzyme's activity, negatively affecting viral replication [36]. Although direct studies on aspartic acid's role in HRV are lacking, these findings from related picornaviruses suggest that preferential use of GAT may contribute to immune evasion by promoting a cellular environment more favorable to viral persistence and less responsive to antiviral signaling.

CTG, which codes for leucine, was among the most underused codons in both the HRV and WNV genomes compared to humans (**Figure 2B-C**). Interestingly, CTC, which also codes for leucine, was found to be equally represented across HRV, WNV, and human genomes. The selective underuse of CTG, but not other leucine codons like CTC, suggests that viral codon usage is shaped more by translational and structural constraints than by amino acid composition alone—potentially reflecting optimization for host tRNA pools, avoidance of RNA secondary structures, or reduced competition with host protein synthesis.

A similar codon-specific pattern was observed for arginine, which is essential to the life cycle of many RNA viruses, so much that its therapeutic depletion has been explored as a treatment for COVID-19 [37]. Out of 6 codons that code for arginine, only AGA was overrepresented in HRV. In contrast, 3 of the 6 arginine codons are overrepresented in WNV despite its status as a dead-end infection in humans. This suggests that codon frequency alone may not be a reliable indicator of functional importance and underscores the need to integrate codon analysis with functional studies to better understand amino acid usage in viral life cycles.

The average CAI value for WNV was found to be significantly higher than that of HRV (**Figure 2D**), which was unexpected considering that WNV is a mosquito-borne virus from the *Flavivirus* family while HRV is a human virus from the *picornavirus* family. Consequently, HRV is known to have evolved to interact with and replicate in the human host. In contrast, WNV has evolved to infect animals like birds and replicate in mosquitoes, making it less adapted to humans than HRV [38]. WNV's higher CAI therefore challenges the notion that CAI is a reliable indicator of overall viral adaptation to a particular host.

Our findings align with recent studies suggesting that codon usage bias in RNA viruses is shaped more by mutational pressure—particularly nucleotide and dinucleotide composition—than by translational selection based on host tRNA availability [39]. In the case of HRV, previous research has shown that CpG suppression, which was discussed earlier as a well-documented feature in many RNA viruses, occurs independently of codon optimization [40]. This implies that viral genomes may prioritize evolving to avoid immune recognition rather than to maximize translational efficiency. Since both HRV and WNV are positive-strand RNA viruses, it's likely

that their codon usage patterns may be driven by intrinsic genomic constraints rather than adaptation to human translational machinery. In other words, WNV's codon usage might theoretically better match that of the human genome, but CAI doesn't account for other crucial evolutionary pressures like CpG suppression, RNA structure, dinucleotide base frequencies, and host immune recognition. In contrast, HRV may exhibit a lower CAI by, for example, actively avoiding CpG-rich codons and favoring A/T nucleotides to evade host immune detection, even at the cost of reduced translation efficiency [41].

Subdued Innate Immune Response to Rhinovirus Infection Reflects Host-Adapted Immune Evasion Strategy

A comparison between the top induced genes in HRV and WNV revealed a contrast in the innate immune responses that were triggered in human macrophages (**Figure 3A**). While both viruses induced upregulation of antiviral genes, HRV infection primarily led to the induction of classic interferon-stimulated genes (ISGs), which are typically upregulated downstream of interferon signaling (**Table 1**). Typical functions of ISGs include blocking viral replication, enhancing antigen presentation, and recruiting immune cells [42]. In contrast, WNV infection resulted in the expression of Type I interferons themselves, which are upstream cytokines that activate a broader antiviral pathway, including the expression of ISGs [43]. This distinction suggests that WNV may be more effectively sensed by innate immune pathways, leading to direct activation of interferon production and, in turn, a broader antiviral signaling pathway. The observed ISG expression in HRV infected cells may reflect a lower level of interferon signaling or delayed activation rather than a robust, upstream immune response.

To further probe innate immune activation, we examined expression levels of four key genes involved in sensing single-stranded RNA viruses: ZC3HAV1, RIG-I, TLR3, and TLR8 (**Figure 3B**). Each of these sensors plays a distinct role in recognizing viral RNA, and all were more strongly upregulated in WNV infection than in HRV. ZC3HAV1, which encodes the ZAP protein, was particularly notable in the context of CpG suppression [44]. As discussed earlier, both WNV and HRV genomes show low CpG content, but this suppression is especially pronounced in HRV, which is well adapted to replication in human hosts. Since ZAP preferentially targets unmethylated CpG-rich RNA for degradation, reduced expression of ZC3HAV1 during HRV infection may reflect the virus's ability to evade ZAP-mediated sensing. In contrast, WNV, a dead-end host virus in humans, induces stronger ZC3HAV1 expression—likely because it does not suppress CpG to the same extent, leaving it more exposed to this sensing mechanism. This finding reaffirms that CpG suppression is a sign of a successful evolutionary adaptation that contributes to immune evasion.

WNV's higher expression of RIG-I, TLR3, and TLR8 further supports this trend. RIG-I is a cytoplasmic sensor that detects viral RNA and triggers type I interferon production [45], while TLR3 and TLR8 are endosomal Toll-like receptors that recognize single-stranded viral RNA, including unmethylated CpG sites [46,47]. The consistently stronger induction of these genes in

WNV-infected macrophages further suggests greater activation of upstream innate sensing pathways. In contrast, their relatively muted expression in HRV-infected cells—though still upregulated—may reflect the virus’s ability to evade early immune detection, further highlighting its effectiveness at evading the human innate immune system.

In addition to examining select genes, pathway enrichment provided a broader view of how each virus engages host immune defenses. While both HRV and WNV activated key innate immune pathways, WNV infection consistently showed stronger enrichment across all eight pathways analyzed (**Figure 3C**). Scores were particularly elevated for pathways regulating defense response to viruses, viral life cycles, and viral genome replication. This is consistent with our previous findings showing that WNV triggers a more comprehensive antiviral response at the systems level. In contrast, the comparatively weaker enrichment during HRV infection reinforces its ability to evade the immune system—not by avoiding pathway activation completely, but by dampening or delaying it. This difference in enrichment patterns may serve as a useful metric for identifying the effectiveness of viral immune evasion strategies, with broader and more intense enrichment indicating failed evasion and more limited host adaptation.

Persistent Human Selective Pressures May Underlie Greater Sequence Divergence in Rhinovirus
Analysis of sequence divergence revealed that HRV accumulates genetic differences at an average rate of 1.22×10^{-2} nucleotide differences per base per year, nearly two orders of magnitude higher than the 1.07×10^{-4} nucleotide differences per base per year observed for WNV (**Figure 4A**). This stark contrast underscores the highly dynamic and rapidly evolving nature of HRV compared to WNV. The greater sequence divergence of HRV points to both intrinsic molecular mechanisms and external selective pressures driving its rapid genetic diversification.

A major driver of this more rapid divergence in HRV appears to be immune-mediated and therapeutic selection, which is markedly stronger in HRV than in WNV. Unlike WNV, which primarily circulates in mosquito–bird transmission cycles and only occasionally infects humans, HRV spreads exclusively among humans and is responsible for widespread, repeated infections throughout life and faces intermittent therapeutic pressures. While no FDA-approved antivirals currently exist for HRV, investigational treatments—such as intranasal interferon and capsid-binding compounds—have been trialed in both clinical and experimental settings [48]. These interventions, though not widespread, can exert strong localized selective pressure, particularly on surface-exposed capsid regions targeted by both host antibodies and antiviral compounds.

In contrast, WNV is a zoonotic virus that primarily cycles between birds and mosquitoes, with humans representing dead-end hosts who are rarely involved in onward transmission. As such, WNV infections in humans are sporadic and typically not subjected to frequent antiviral

treatment, meaning the virus faces minimal immune or therapeutic selection in human hosts. Its evolutionary trajectory is therefore shaped more by ecological constraints and vector-host dynamics than by ongoing pressure to evade the human immune system.

The sustained exposure of HRV to both host immunity and targeted therapeutic agents creates a dynamic and selective environment that favors variants capable of antigenic drift and immune escape. Supporting this, recent *in vitro* evolution studies have shown that under selective conditions—such as capsid-destabilizing environments—HRV rapidly acquires mutations at key antigenic sites [49]. This reinforces the idea that immune and therapeutic evasion are central forces driving HRV's genetic diversification over time.

Phylogenetic reconstruction and UMAP visualizations both revealed clear temporal clustering among HRV strains, with earlier strains localized near the root and more recent strains diverging into distinct clades (**Figure 4B-C**). Although our analysis focused on the US, similar spatio-temporal patterns have been documented in other geographic regions, where repeated viral introductions and localized expansions of specific genotypes contribute to shifting patterns of dominance over time [50]. These patterns are consistent with a virus undergoing rapid evolution in response to persistent immune and environmental pressures.

The observed rate of sequence divergence, coupled with a strong temporal structure, presents significant challenges for the development of broadly protective therapies and vaccines. Antigenic drift can quickly undermine interventions, especially those targeting highly variable regions. Ongoing genomic surveillance, including integrating global sequence data with antigenic and immune profiling, will be essential for identifying conserved targets and anticipating emerging variants, ultimately guiding the design of more durable, population-level interventions [51].

CONCLUSION

In our study, we investigated the multi-layered immune evasion strategies of HRV and compared these to WNV, a dead-end host in humans. Our findings reveal that HRV employs a coordinated set of molecular tactics to evade innate immune detection and sustain replication in human hosts.

We observed that CpG suppression and TpG enrichment occur in tandem across the HRV genome, particularly in regions encoding surface-exposed capsid proteins like VP1 and VP2. This region-specific pattern suggests a functional prioritization of immune-interacting proteins. The markedly lower CpG levels in HRV—especially when contrasted with WNV—could serve as a quantitative signature of adaptation to the human immune system. Codon usage analysis further showed that HRV's genome is shaped more by immune avoidance than translational optimization, as evidenced by a lower codon adaptation index (CAI) relative to WNV. This

challenges the traditional interpretation of CAI as a marker for host adaptation and instead highlights a trade-off between immune evasion and translation efficiency.

Our transcriptional analysis revealed that HRV induces a more subdued host innate immune response compared to WNV, characterized by lower expression of upstream antiviral cytokines and innate immune sensors. While HRV still activates ISGs, the relatively muted upstream response implies a strategic evasion of early immune detection. Complementary pathway analysis confirmed that WNV infection induces a broader and more robust antiviral response. Finally, HRV exhibited greater genomic variability across strains and over time compared to WNV. This mutational flexibility likely contributes to its antigenic diversity and immune escape, in contrast to the relative stability of WNV, which circulates primarily in non-human reservoirs and faces limited exposure to human immune responses and antiviral treatments.

Together, these findings—region-specific CpG suppression and TpG enrichment, codon bias, immune-muted host responses, and higher mutability—form a complementary and interconnected framework of immune evasion. Rather than acting independently, these mechanisms reinforce each other: for instance, CpG suppression both decreases recognition by host sensors and shapes codon usage, while mutation-driven antigenic variation enhances the effectiveness of evasion tactics by continually evolving the viral genome recognized by the immune system and antiviral pathways.

Our analysis was constrained by several limitations. For genomic analysis, we were limited to 16 HRV strains and 12 WNV strains, which may not capture the full genetic diversity of either virus. Expanding the number and variety of strains could help generalize our findings and uncover more nuanced patterns of immune evasion. Additionally, we performed genomic analyses primarily on HRV-C strains, while our transcriptional host response data came from HRV-A16 infections. This inconsistency may introduce species-specific biases that confound the interpretation of immune evasion strategies. Host responses were also measured using different platforms—microarray for HRV and RNAseq for WNV—limiting the comparability due to differences in sensitivity and dynamic range. Ideally, both infections should be performed on the same donor under identical conditions using RNA-seq to ensure accurate comparison of host responses.

Our codon usage analysis also has important caveats. CAI values were calculated using average human codon frequencies, which may not reflect the specific codon preferences of respiratory epithelial cells—the primary targets of HRV. This may reduce the accuracy of CAI as a metric for translational adaptation in this context. Examining other factors—such as mRNA secondary structure, tRNA abundance, and the tRNA Adaptation Index—could provide a more complete picture of how codon usage influences viral protein synthesis and immune evasion.

Notably, our functional enrichment analysis was also limited to manually selected GO terms with pathway overlap, which may introduce interpretive bias and limit discovery of novel or unexpected pathways. A more systematic approach may yield a more comprehensive view of viral impact on host processes. Additionally, mutation rates for each virus were calculated from different time spans and with unequal strain representation between HRV and WNV, which may skew the results and limit conclusions about evolutionary dynamics. Finally, while WNV served as a useful contrast due to its phylogenetic distance and divergent host ecology, its rarity and zoonotic lifestyle could introduce confounding variables. A more appropriate comparison for future studies might be another respiratory virus like seasonal influenza.

Our findings open several avenues for future research. A quantitative study across multiple viruses could validate CpG suppression as a broad marker of immune evasion strength. Similarly, systematic correlation analyses between CpG suppression and TpG enrichment could clarify whether they are consistently anticorrelated and functionally linked across viral taxa. Additionally, an expanded temporal and strain-based mutation analysis could help discern whether rapid mutation is a generalizable evasion strategy or specific to HRV. Future work can also adopt unsupervised or machine learning approaches for pathway enrichment and host response profiling, which could reveal previously unrecognized immune signatures or interactions. Finally, integrating transcriptional response data with genome-wide association studies (GWAS) could help identify host genetic variants that influence susceptibility to HRV infection and disease severity. Top DEGs with known immune functions could serve as promising candidates for future GWAS aimed at uncovering genetic determinants of antiviral defense.

By integrating metrics such as CpG suppression levels, TpG enrichment, codon adaptation index deviations, transcriptional immune response profiles, and mutation rates, researchers could generate a single, scalable score to predict a virus's immune evasion strength. Such a tool would have profound implications for public health, allowing for the rapid classification of emerging pathogens based on their immune evasion potential. It could also inform vaccine design strategies, helping researchers prioritize targets that are less prone to evasion mechanisms. Ultimately, our study underscores the need to shift from pathogen-specific to mechanism-based frameworks in antiviral research, paving the way for interventions that can outmaneuver even the most evasive viruses.

REFERENCES

1. Stanway G, Hughes PJ, Mountford RC, Minor PD, Almond JW. The complete nucleotide sequence of a common cold virus: human rhinovirus 14. *Nucleic Acids Res.* 1984;12(20):7859-7875. doi:10.1093/nar/12.20.7859
2. Jacobs SE, Lamson DM, St George K, Walsh TJ. Human Rhinoviruses. *Clin Microbiol Rev.* 2013;26(1):135-162. doi:10.1128/CMR.00077-12
3. Hershenson MB. Rhinovirus-induced exacerbations of asthma and COPD. *Scientifica (Cairo).* 2013;2013:405876. doi:10.1155/2013/405876
4. Glanville N, Johnston SL. Challenges in developing a cross-serotype rhinovirus vaccine. *Curr Opin Virol.* 2015;11:83-88. doi:10.1016/j.coviro.2015.03.004
5. Lee WM, Lemanske RF, Evans MD, et al. Human rhinovirus species and season of infection determine illness severity. *Am J Respir Crit Care Med.* 2012;186(9):886-891. doi:10.1164/rccm.201202-0330OC
6. Chan YK, Gack MU. Viral evasion of intracellular DNA and RNA sensing. *Nat Rev Microbiol.* 2016;14(6):360-373. doi:10.1038/nrmicro.2016.45
7. Walsh D, Mohr I. Viral subversion of the host protein synthesis machinery. *Nat Rev Microbiol.* 2011;9(12):860-875. doi:10.1038/nrmicro2655
8. Bonjardim CA, Ferreira PCP, Kroon EG. Interferons: signaling, antiviral and viral evasion. *Immunol Lett.* 2009;122(1):1-11. doi:10.1016/j.imlet.2008.11.002
9. Drake JW, Holland JJ. Mutation rates among RNA viruses. *Proc Natl Acad Sci U S A.* 1999;96(24):13910-13913. doi:10.1073/pnas.96.24.13910
10. Laperriere V, Brugger K, Rubel F. Simulation of the seasonal cycles of bird, equine and human West Nile virus cases. *Prev Vet Med.* 2011;98(2):99-110. doi:10.1016/j.prevetmed.2010.10.013
11. Benson DA, Clark K, Karsch-Mizrachi I, Lipman DJ, Ostell J, Sayers EW. GenBank. *Nucleic Acids Res.* 2014;42(Database issue):D32-D37. doi:10.1093/nar/gkt1030
12. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. Basic local alignment search tool. *J Mol Biol.* 1990;215(3):403-410. doi:10.1016/S0022-2836(05)80360-2
13. Huber W, Carey VJ, Gentleman R, et al. Orchestrating high-throughput genomic analysis with Bioconductor. *Nat Methods.* 2015;12(2):115-121. doi:10.1038/nmeth.3252.
14. Jansen R, Bussemaker HJ, Gerstein M. Revisiting the codon adaptation index from a whole-genome perspective: analyzing the relationship between gene expression and codon occurrence in yeast using a variety of models. *Nucleic Acids Res.* 2003;31(8):2242-2251. doi:10.1093/nar/gkg306

15. Edgar R, Domrachev M, Lash AE. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res.* 2002;30(1):207-210. doi:10.1093/nar/30.1.207
16. Kawai T, Akira S. Innate immune recognition of viral infection. *Nat Immunol.* 2006;7(2):131-137. doi:10.1038/ni1303
17. Robinson MD, Speed TP. A comparison of Affymetrix gene expression arrays. *BMC Bioinform.* 2007;8(1):449. doi:10.1186/1471-2105-8-449
18. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* 2014;15(12):550. doi:10.1186/s13059-014-0550-8
19. Carvalho BS, Irizarry RA. A framework for oligonucleotide microarray preprocessing. *Bioinformatics.* 2010;26(19):2363-2367. doi:10.1093/bioinformatics/btq431
20. Ritchie ME, Phipson B, Wu D, et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* 2015;43(7):e47. doi:10.1093/nar/gkv007
21. Stelzer G, Rosen N, Plaschkes I, et al. The GeneCards Suite: from gene data mining to disease genome sequence analyses. *Curr Protoc Bioinformatics.* 2016;54:1.30.1-1.30.33. doi:10.1002/cpbi.5
22. Xu S, Hu E, Cai Y, et al. Using clusterProfiler to characterize multiomics data. *Nat Protoc.* 2024;19(11):3292-3320. doi:10.1038/s41596-024-01020-z
23. Wright ES. DECIPHER: harnessing local sequence context to improve protein multiple sequence alignment. *BMC Bioinform.* 2015;16:322. doi:10.1186/s12859-015-0749-z
24. Schliep KP. phangorn: phylogenetic analysis in R. *Bioinformatics.* 2011;27(4):592-593. doi:10.1093/bioinformatics/btq706
25. McInnes L, Healy J, Melville J. UMAP: Uniform manifold approximation and projection for dimension reduction. *J Open Source Softw.* 2019;3(29):861. doi:10.21105/joss.00861
26. Piovesan A, Pelleri MC, Antonaros F, Strippoli P, Caracausi M, Vitale L. On the length, weight and GC content of the human genome. *BMC Res Notes.* 2019;12(1):106. doi:10.1186/s13104-019-4137-z
27. Greenbaum BD, Levine AJ, Bhanot G, Rabadan R. Patterns of evolution and host gene mimicry in influenza and other RNA viruses. *PLoS Pathog.* 2008;4(6):e1000079. doi:10.1371/journal.ppat.1000079
28. Xia X. Extreme genomic CpG deficiency in SARS-CoV-2 and evasion of host antiviral defense. *Mol Biol Evol.* 2020;37(9):2699-2705. doi:10.1093/molbev/msaa094

29. Upadhyay M, Samal J, Kandpal M, et al. CpG dinucleotide frequencies reveal the role of host methylation capabilities in parvovirus evolution. *J Virol.* 2013;87(24):13816-13824. doi:10.1128/JVI.02515-13
30. Krieg AM. Immune effects and mechanisms of action of CpG motifs. *Vaccine.* 2000;19(6):618-622. doi:10.1016/S0264-410X(00)00249-8
31. Lin YT, Chiweshe S, McCormick D, et al. Human cytomegalovirus evades ZAP detection by suppressing CpG dinucleotides in the major immediate early 1 gene. *PLoS Pathog.* 2020;16(9):e1008844. doi:10.1371/journal.ppat.1008844
32. Ohno S. Universal rule for coding sequence construction: TA/CG deficiency - TG/CT excess. *Proc Natl Acad Sci U S A.* 1988;85(24):9630-9634. doi:10.1073/pnas.85.24.9630
33. Verdaguer N, Blaas D, Fita I. Structure of human rhinovirus serotype 2 (HRV2). *J Mol Biol.* 2000;300(5):1179-1194. doi:10.1006/jmbi.2000.3943
34. Wu G. Amino acids: metabolism, functions, and nutrition. *Amino Acids.* 2009;37(1):1-17. doi:10.1007/s00726-009-0269-0
35. Liu H, Zhu Z, Xue Q, et al. Picornavirus infection enhances aspartate by the SLC38A8 transporter to promote viral replication. *PLoS Pathog.* 2023;19(2):e1011126. doi:10.1371/journal.ppat.1011126
36. Jablonski SA, Morrow CD. Mutation of the aspartic acid residues of the GDD sequence motif of poliovirus RNA-dependent RNA polymerase results in enzymes with altered metal ion requirements for activity. *J Virol.* 1995;69(3):1532-1539. doi:10.1128/JVI.69.3.1532-1539.1995
37. Grimes JM, Khan S, Badeaux M, Rao RM, Rowlinson SW, Carvajal RD. Arginine depletion as a therapeutic approach for patients with COVID-19. *Int J Infect Dis.* 2021;102:566-570. doi:10.1016/j.ijid.2020.10.100
38. Sejvar JJ. West Nile virus: an historical overview. *Ochsner J.* 2003;5(3):6-10.
39. Belalov IS, Lukashev AN. Causes and implications of codon usage bias in RNA viruses. *PLoS One.* 2013;8(2):e56642. doi:10.1371/journal.pone.0056642
40. Megremis S, Demetriou P, Makrinioti H, Manoussaki AE, Papadopoulos NG. The genomic signature of human rhinoviruses A, B and C. *PLoS One.* 2012;7(9):e44557. doi:10.1371/journal.pone.0044557
41. Khandia R, Singhal S, Kumar U, et al. Analysis of Nipah virus codon usage and adaptation to hosts. *Front Microbiol.* 2019;10:886. doi:10.3389/fmicb.2019.00886
42. Doly J, Civas A, Navarro S, Uze G. Type I interferons: expression and signalization. *Cell Mol Life Sci.* 1998;54(10):1109-1121. doi:10.1007/s000180050240

43. Ivashkiv LB, Donlin LT. Regulation of type I interferon responses. *Nat Rev Immunol*. 2014;14(1):36-49. doi:10.1038/nri3581
44. Lee H, Komano J, Saitoh Y, et al. Zinc-finger antiviral protein mediates retinoic acid-inducible gene I-like receptor-independent antiviral response to murine leukemia virus. *Proc Natl Acad Sci U S A*. 2013;110(30):12379-12384. doi:10.1073/pnas.1310604110
45. Yoneyama M, Fujita T. Recognition of viral nucleic acids in innate immunity. *Rev Med Virol*. 2010;20(1):4-22. doi:10.1002/rmv.633
46. Tatematsu M, Nishikawa F, Seya T, Matsumoto M. Toll-like receptor 3 recognizes incomplete stem structures in single-stranded viral RNA. *Nat Commun*. 2013;4(1):1833. doi:10.1038/ncomms2857
47. Kell AM, Gale M. RIG-I in RNA virus recognition. *Virology*. 2015;479-480:110-121. doi:10.1016/j.virol.2015.02.017
48. Farr BM, Gwaltney JM, Adams KF, Hayden FG. Intranasal interferon-alpha 2 for prevention of natural rhinovirus colds. *Antimicrob Agents Chemother*. 1984;26(1):31-34. doi:10.1128/aac.26.1.31
49. Murer L, Petkidis A, Vallet T, Vignuzzi M, Greber UF. Chemical evolution of rhinovirus identifies capsid-destabilizing mutations driving low-pH-independent genome uncoating. *J Virol*. 2022;96(2):e01060-21. doi:10.1128/JVI.01060-21
50. Goya S, Wendt ST, Xie H, et al. Genomic epidemiology and evolution of rhinovirus in western Washington state, 2021–2022. *medRxiv* [Preprint]. 2024 Feb 14:2024.02.13.24302773. doi:10.1101/2024.02.13.24302773.
51. Kistler AL, Webster DR, Rouskin S, et al. Genome-wide diversity and selective pressure in the human rhinovirus. *Virol J*. 2007;4(1):40. doi:10.1186/1743-422X-4-40

DATA AVAILABILITY

Genomic Signatures and Codon Usage Bias

Data Resource	GenBank Accession Number	
Complete viral genome FASTA files with feature information	Rhinovirus-C (16 Strains) JN205461, JX074056, KY348786, MZ268694, EF582387, NC_009996, JF907574, MZ667421, KF734978, KJ675507, MK989757, MK989751, MF806525, OK161404, PV178494, PV178425	West Nile Virus (12 strains) NC_009942, EF571854, KC407666, EF571854, AF404754, AF260967, HQ596519, EF657887, AF404756, EF530047, KJ501530, DQ164202

Human RNAseq and Microarray Data:

Data Resource	GEO Accession Number
Microarray data Human macrophages (Rhinovirus-A16)	GSE106963
RNAseq Human macrophages (West Nile Virus)	GSE40718

Evolutionary Divergence Analysis

Data Resource	GenBank Accession Number	
Complete viral genome FASTA files with feature information	Rhinovirus-C (54 US strains between 2016 and 2024) KY348786, KY369878, MK520815, OQ116581, ON311156, ON311180, MZ268691, MZ460691, OK017949, OK649386, ON311177, ON311176, ON311179, ON311175, MW969524, MW969527, MW969536, ON311155, MZ363453, PP158745, MZ268717, MZ268706, MZ322915, MZ363437, MZ427505, OL961524, OK017918, OL961531, MZ629145, MZ670576, OK161367, OK161371, OK161404, OK539509, PP194066, PP187406, PV178322, PP314216, PV178503, PV178487, PV178494, PV178470, PV178551, PV178585, PV178348, PV178619, PV178458,	West Nile Virus (79 US strains between 1999 and 2008) FJ151394, HQ596519, MZ605381, EF530047, EF657887, MH166882, MH166901, MH166903, MH166904, DQ164194, KJ501234, KJ501244, KJ501343, KJ501363, KJ501395, KJ501439, KJ501443, KJ501448, KJ501468, KJ501480, KJ501486, KJ501496, KJ501516, KJ786934, MH166886, MH170273, DQ164186, DQ164193, DQ164195, DQ164197, DQ164198, DQ164200, DQ164205, KJ501263, KJ501279, KJ501290, KJ501296, KJ501298, KJ501302, KJ501310, KJ501315, KJ501318, KJ501322, KJ501344, KJ501370, KJ501371, KJ501373,

	PV178608, PV178548, PV178386, PV178591, PV178425, PV178222, PV17820	KJ501380, KJ501384, KJ501399, KJ501466, KJ501474, KJ501475, KJ501483, KJ501510, KJ501511, MH166883, MH166905, MH170275, DQ164188, DQ164191, KJ501456, MH170234, HM488144, KJ501237, MH166892, GQ379156, AF196835, AF260967, AF404753, AF404754, AF404755, AF404756, AF533540, AY646354, AY795965, AY842931, EF571854, NC_009942
--	---	---

CODE AVAILABILITY

All R code developed and used for project can be found at the following link:

<https://github.com/wuoli/biohackathon2025-rhinovirus>

FIGURES AND TABLES

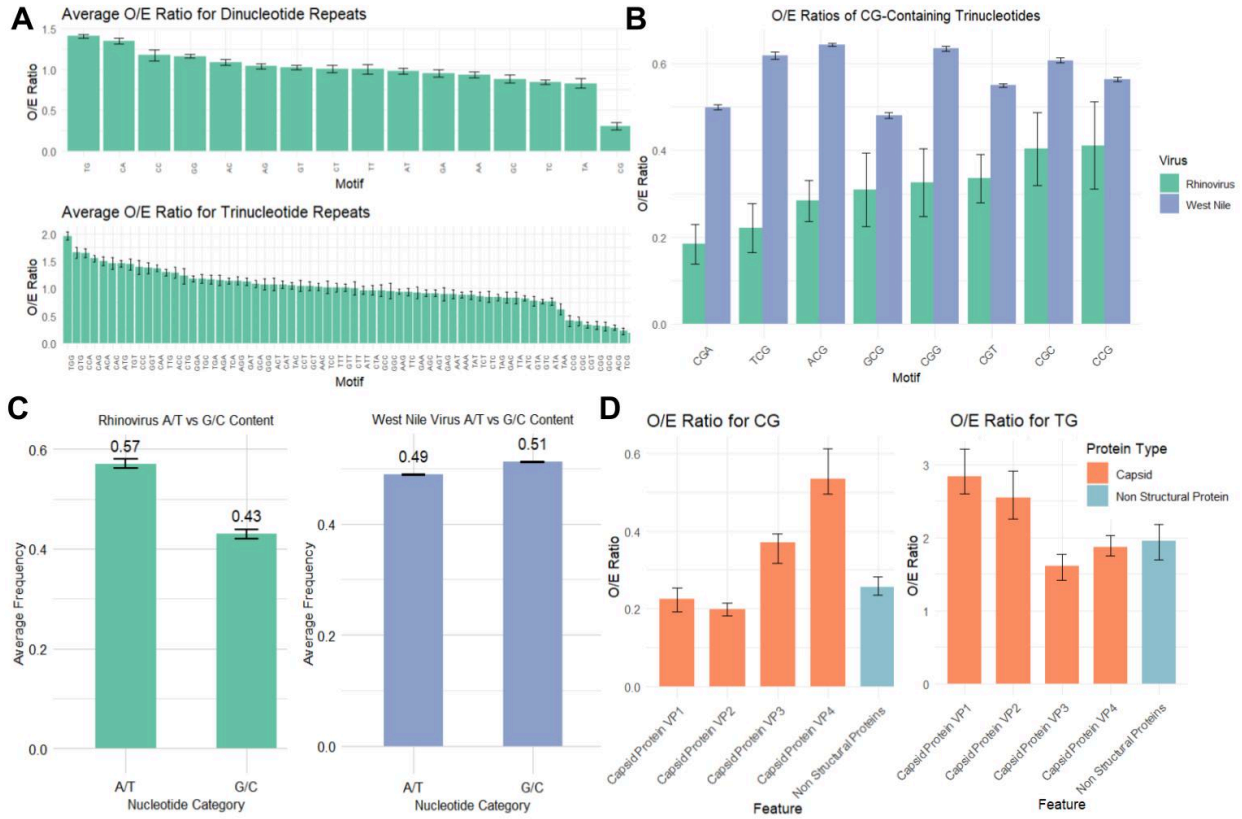


FIGURE 1: *Rhinovirus Genome is Defined by Region-Specific Underrepresentation of CpG and Enrichment of TpG.*

- Bar chart (with error bars) of average observed/expected (O/E ratio) across 16 Rhinovirus strains for each dinucleotide and trinucleotide motif in its genome.
- Bar chart with error bars of average O/E ratio of CpG-containing trinucleotide motif for Rhinovirus (16 strains) and West Nile Virus (12 strains).
- Bar chart with error bars of average O/E ratio of each CpG-containing trinucleotide motif for Rhinovirus (16 strains) and West Nile Virus (12 strains).
- Bar chart (with error bars) of average O/E ratio for CpG and TpG dinucleotides for each feature-coding region across Rhinovirus strains.

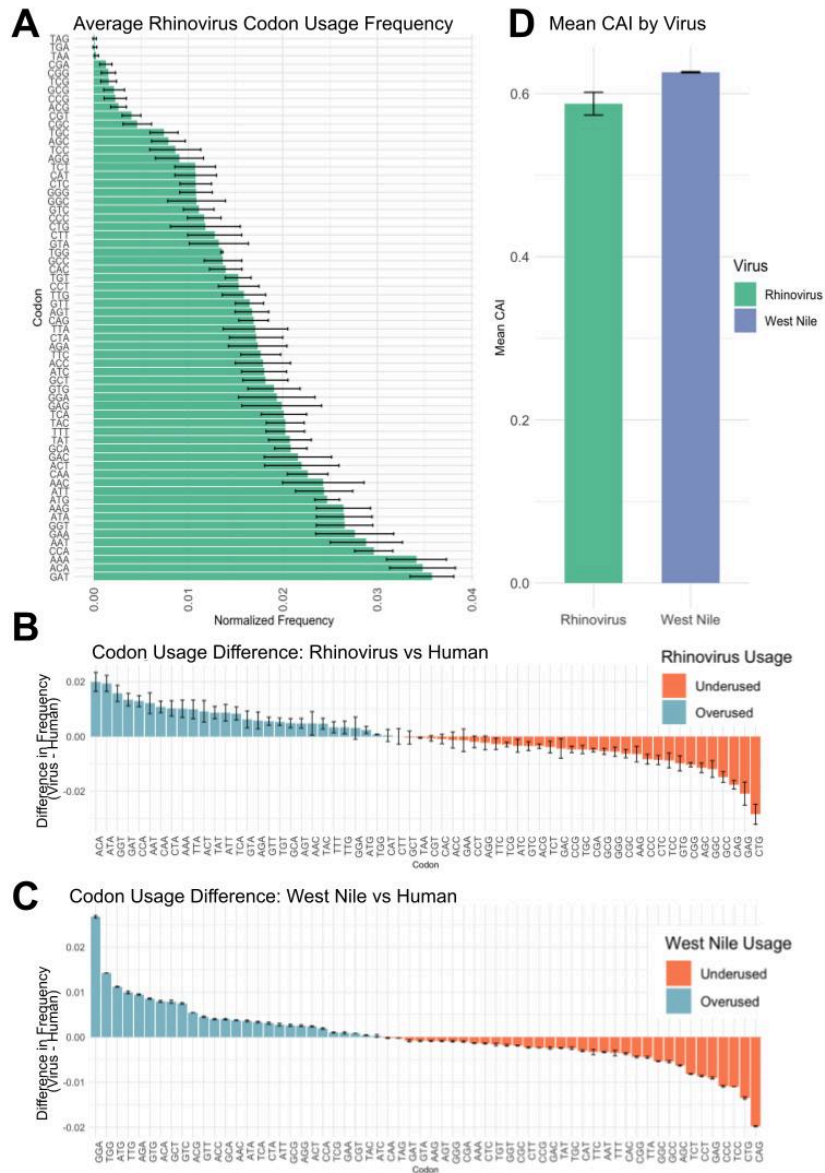


FIGURE 2: *West Nile Virus Exhibits Greater Codon Adaptation Than Rhinovirus Despite Human Host Specialization.*

- Bar chart (with error bars) of the average frequency of each codon across 16 Rhinovirus strains.
- Mean difference chart (with error bars) of the difference in codon usage frequency between the Rhinovirus and human genomes (Rhinovirus - human).
- Mean difference chart (with error bars) of the difference in codon usage frequency between the West Nile Virus and human genomes (West Nile - human).
- Bar chart (with error bars) of average CAI values across the 16 Rhinovirus strains and 12 West Nile strains.

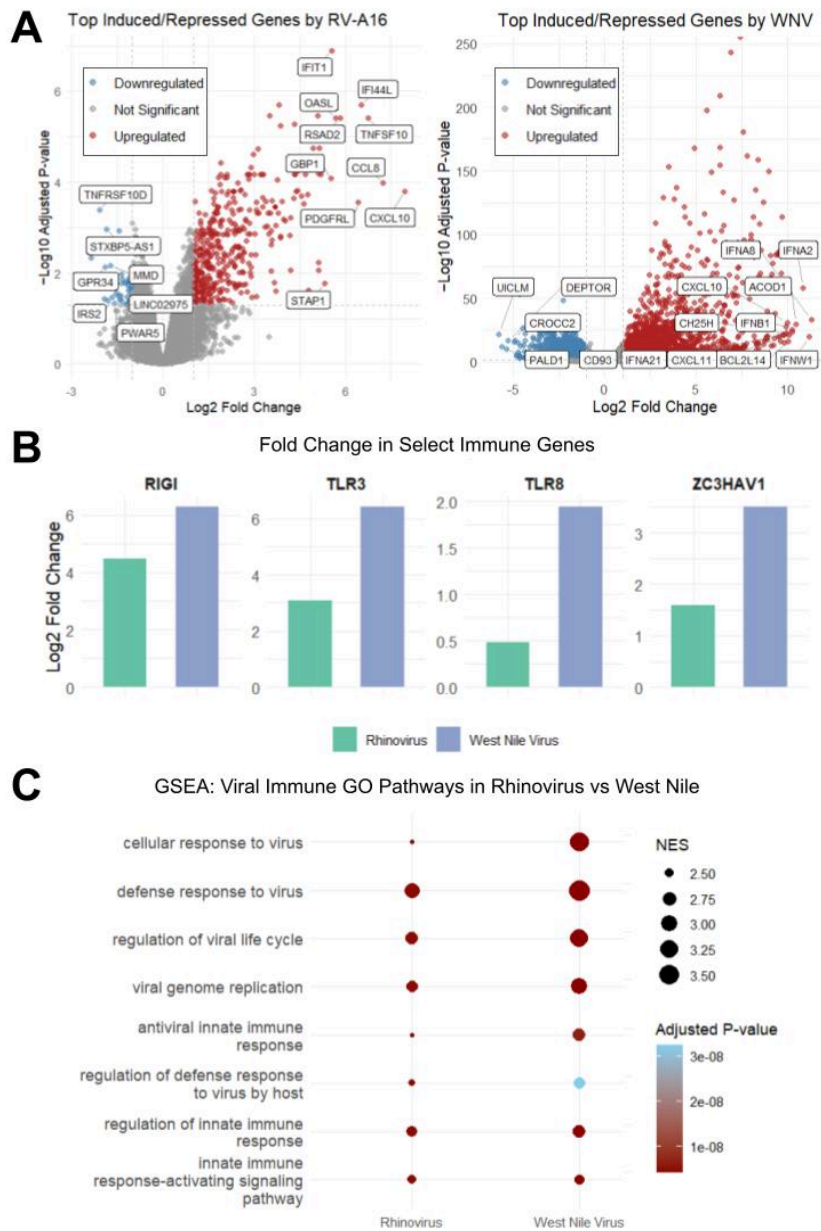


FIGURE 3: *West Nile Virus Induces More Extensive Activation of Antiviral and Viral Sensing Pathways Than Rhinovirus.*

- Volcano plots of significant differentially expressed genes ($\log_2FC > 1$, adjusted p-value < 0.05) following Rhinovirus-A16 and West Nile Virus infection in human macrophages.
- Bar graphs of Rhinovirus and West Nile Virus $\log_2FC$ s for genes associated with single-stranded RNA viruses sensing: RIGI, TLR3, TLR8, and ZC3HAV1.
- Gene Set Enrichment Analysis for select viral and immune-related Gene Ontology biological pathway terms showing (1) normalized enrichment scores (NES) and (2) adjusted p-values.

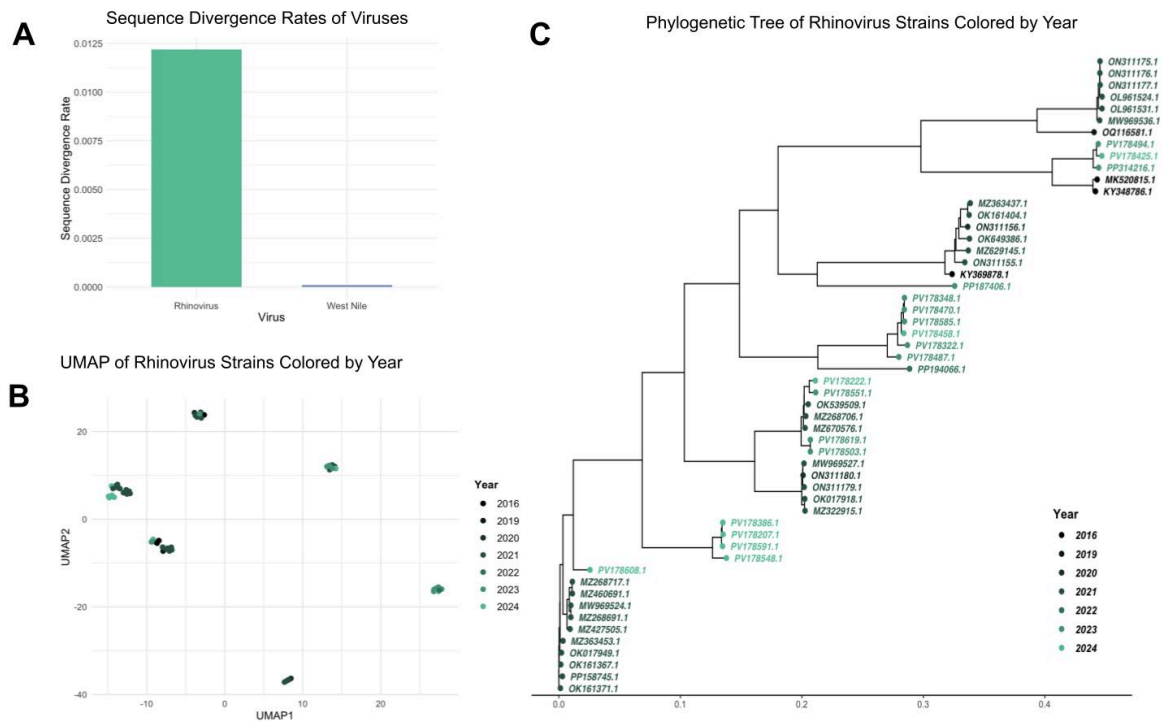


FIGURE 4: *Rhinovirus Shows Greater Sequence Divergence Rates than West Nile Virus.*

- A. Bar plot comparing the estimated divergence rates (average number of mutations accumulated per base per year) across 54 Rhinovirus strains (2016–2024) and 79 West Nile Virus strains (1999–2008) collected in the US.
- B. UMAP projection of the 54 Rhinovirus strains colored by collection year.
- C. Maximum Likelihood phylogenetic tree of the 54 Rhinovirus strains using a Neighbor-Joining tree as base model. Accession numbers from NCBI are shown for each strain, and tips are colored by year.

TABLE 1: Functions of Top Upregulated and Downregulated Genes Following Rhinovirus/West Nile Virus Infection (sourced from GeneCards [21])

Gene	Virus	Regulation	Function	Phenotype Associated	Source
IFIT1	HRV	up	RNA-binding antiviral protein; inhibits non-self viral RNA translation; involved in innate immune recognition.	Increases VACV infection, IL-8 secretion, NF- κ B activation; affects viability and immune response.	Link
OASL	HRV	up	Binds dsRNA; antiviral activity via RNase L-independent pathway.	Affects viability, cholesterol metabolism, and immune function.	Link
IFI44L	HRV	up	IFN-stimulated gene; antiviral and antibacterial; regulates cytokine secretion and IFN feedback.	Affects NF- κ B expression, DNA content, hepcidin mRNA expression, and viral infection response.	Link
TNFRSF10A	HRV	up	TRAIL receptor inducing apoptosis via caspase activation; activates NF- κ B.	Impacts metabolism, immune system, and mortality; involved in apoptosis regulation.	Link
CCL8	HRV	up	Chemotactic factor attracting monocytes, lymphocytes, eosinophils; may modulate inflammation and neoplasia; binds heparin.	Increased gamma-H2AX, decreased vaccinia virus infection, altered viability, changes in endosome-nucleus distance.	Link
RSAD2	HRV	up	Interferon-induced antiviral protein; converts CTP to ddhCTP to inhibit viral RNA polymerases; affects T-cell activation and protein secretion.	Increased vaccinia virus infection, immune and metabolic alterations, decreased viability, increased IL-8 secretion.	Link
GBP1	HRV	up	IFN-inducible GTPase; promotes pathogen detection, inflammasome activation, and autophagy; protects against various pathogens.	Increased gamma-H2AX, decreased viability, Wnt pathway downregulation, synthetic lethality with vaccinia virus.	Link
CXCL10	HRV	up	Pro-inflammatory cytokine; regulates immune cell movement and activation; crucial in viral infections and brain injury response.	Changes in viability, immune and nervous system phenotypes, synthetic lethality with c-Myc.	Link
PDGFRL	HRV	up	Encodes protein similar to the ligand binding domain of platelet-derived growth factor receptor beta; May function as a tumor suppressor	Increased VACV infection, decreased HPV/influenza replication, synthetic lethality with NAE inhibitor MLN4924, variable viability.	Link
STAP1	HRV	up	Docking protein in BCR signaling; enhances TEC activity in positive feedback loop.	Altered cell viability and proliferation, increased EGFR surface abundance, reproductive system phenotype.	Link
TNFRSF10D	HRV	down	TRAIL receptor; protects against TRAIL-induced apoptosis; possible NF- κ B pathway involvement.	Affects viability, IL-8 secretion, viral infection sensitivity, and several systemic phenotypes.	Link
STXBP5-AS1	HRV	down	Primarily functions as a tumor suppressor, inhibiting cell proliferation and invasion in various cancers	Increased susceptibility to vaccinia virus infection.	Link
MMD	HRV	down	Regulates lysosomal membrane dynamics in microglia after brain lesions.	Resistance to VACV infection; impacts on metabolism, growth, immune, neurological, and other systems.	Link
GPR34	HRV	down	GPCR for lysophosphatidylserine; mediates immune response, tissue repair, microglial function.	Altered IL-8 secretion, Nanog expression, and response to VACV; affects immune and nervous systems.	Link
IRS2	HRV	down	Signal transducer for insulin/IGF receptors;	Alters viability, cell migration, Wnt	Link

Gene	Virus	Regulation	Function	Phenotype Associated	Source
			regulates metabolism, growth, Wnt signaling, and mitosis.	signaling; broad systemic effects including endocrine and neurological systems.	
PWAR5	HRV	down	Regulating cell growth and potentially in the development of glioma. Studies suggest that PAR5 inhibits the proliferation, invasion, and migration of human glioma cells	Associated with phf-tau measurement.	Link
LINC02975	HRV	down	Non-protein coding RNA gene		Link
IFNA21	WNV	up	Interferon with antiviral effects; stimulates immune enzyme production.	Affects immune, blood, and respiratory systems. Alters viability and virus infection susceptibility.	Link
CXCL11	WNV	up	Chemoattractant for activated T-cells; involved in immune response and CNS inflammation.	Immune and blood phenotypes; affects cell viability and virus responses.	Link
BCL2L14	WNV	up	Promotes apoptosis.	Impacts immune, digestive, and cancer-related traits. Affects virus resistance, cell number, and drug response.	Link
IFNW1	WNV	up	Type I interferon with antiviral properties.	Increases vaccinia virus infection, affects viability, apoptosis, and cellular transport.	Link
IFNB1	WNV	up	Type I interferon regulating immune responses, antiviral activity, tumor suppression, and neuronal health.	Affects nervous system, immune function, metabolism, cancer, and cell viability under infection or drug exposure.	Link
CH25H	WNV	up	Converts cholesterol to 25-hydroxycholesterol; modulates lipid metabolism, immune signaling, and viral restriction.	Affects immune, reproductive, hematopoietic, and endocrine systems. Alters response to vaccinia virus and cell viability.	Link
CXCL10	WNV	up	Pro-inflammatory cytokine; regulates immune cell movement and activation; crucial in viral infections and brain injury response.	Changes in viability, immune and nervous system phenotypes, synthetic lethality with c-Myc.	Link
ACOD1	WNV	up	Produces itaconate to inhibit inflammation and pathogen replication; suppresses succinate dehydrogenase in neurons.	Affects neurological, immune, and hematopoietic systems; modifies viral infection response and cell viability.	Link
IFNA8	WNV	up	Antiviral interferon produced by macrophages; activates antiviral enzymes.	Impacts immune, respiratory, and hematopoietic systems; alters vaccinia virus resistance and cell morphology.	Link
IFNA2	WNV	up	Macrophage-derived antiviral interferon.	Involved in immune, respiratory, and hematopoietic phenotypes; affects virus infection response and gene interactions.	Link
UICLM	WNV	down	An RNA Gene upregulated in colorectal cancer liver metastasis		Link
DEPTOR	WNV	down	Inhibits mTORC1 and mTORC2 to regulate autophagy and cell growth.	Affects metabolism, growth, reproduction, liver, and neoplasm phenotypes. Increases vaccinia virus infection.	Link

Gene	Virus	Regulation	Function	Phenotype Associated	Source
CROCC2	WNV	down	Promoting the expansion of cortical progenitors during brain development, particularly in the amplification of basal progenitors in human corticogenesis	Impacts growth, cardiovascular, immune, blood, and renal systems.	Link
PALD1	WNV	down	Homolog of murine paladin; function not fully characterized.	Affects metabolism, cardiovascular, respiratory, vision, behavior, and cell phenotypes. Increases vaccinia virus infection.	Link
CD93	WNV	down	Cell surface receptor involved in angiogenesis, inflammation, and phagocytosis.	Alters immune and blood cell functions. Increases viral infection susceptibility and affects autophagy and cell viability.	Link

TABLE 2: *Normalized Enrichment Scores (NES) and Differentially Expressed Genes Associated With Each Gene Ontology Term for Rhinovirus and West Nile Virus*

Gene Ontology Term	Virus	NES	Gene
cellular response to virus	HRV	2.409926	IFIH1/OAS1/CXCL10/IFI6/ZC3H12A/IFNB1/JAK2/TRIM6/CCL19/IL6/TLR3/IKBKE
	WNV	3.023427	IFIH1/IFNA1/IFNA13/EXT1/CCL5/OAS1/IFNA2/ZC3H12A/IFNA8/IFI6/CXCL10/IL6/JAK2/IFNB1/ADAR/TLR3/NFKB1/IFNA21/IFNA4/IFNA16/IFNA7/IFNA10/IFNW1/IFNE/IFNA17/IFNL1/IKBKE/FMR1/IFNA6/IFNA14/CCL19/IL12A/IFNA5/IFNL3/IFNL2/LGALS8/DDX3X/HIF1A/JAK1/TRIM6/IFNGR2/SMAD3/ATF2/IFNL4/NLRP3/RIOK3/HSP90AA1/CHUK/IFNAR1/GBF1/MIR146A/ARF1/IRF5
defense response to virus	HRV	2.934275	IFIT1/IFIT3/IFI44L/OASL/RSAD2/OAS2/DTX3L/RIGI/IFIT2/IFIH1/SHFL/ISG15/NT5C3A/ACOD1/HERC5/MX1/IRF7/MX2/OAS1/EIF2AK2/STAT2/BST2/GBP1/IFIT5/PARP9/USP18/ZC3HAV1/PML/STAT1/GBP5/CXCL10/DDX60/OAS3/IFI27/GBP2/IFITM1/TNF/TRIM5/TRIM25/TRIM22/TRIM21/IFI6/APOBEC3G/ZC3H12A/IRF1/IFNB1/UNC93B1/MOV10/CGAS/AIM2/IFI16/DDX60L/NLRC5/DHX58/PLSCR1/ZNFX1/RTP4/TRIM6/IL15/TREX1/TRIM56/IL6/IRF9/APOBEC3F/CD40/TNFAIP3/IFITM2/LILRB1/MYD88/MLKL/GBP7/IL27/TLR3/IKBKE/CASP1
	WNV	3.202632	IFIT3/RIGI/GBP5/HERC5/MX2/IRF1/OASL/APOBEC3B/OAS2/DHX58/USP18/ISG20/APOBEC3G/NT5C3A/MX1/APOBEC3A/GBP2/ZBP1/GBP1/IFITM1/IFIH1/ZC3HAV1/PARP9/AIM2/IFNA1/IFIT2/IFNA13/IFIT1/OAS3/CGAS/RTP4/OAS1/EIF2AK2/IFI27/STAT2/PLSCR1/IFI44L/ISG15/STAT1/IFNA2/TRIM22/PMAIP1/TRIM5/ZNFX1/IFITM3/IFI16/DTX3L/IFIT5/APOBEC3D/TNF/PML/TRIM26/TNF AIP3/DDX60/IFIT1B/RSAD2/GBP3/IL27/CD40/ZC3H12A/IL15/NLRC5/DDX60L/IFNA8/IL23A/SHFL/IFI6/CXCL10/TRIM25/ACOD1/IRF7/IL6/TRIM21/LILRB1/IFNB1/ADAR/CASP1/IRF9/SLFN13/TLR3/NT5C2/IFNA21/TLR2/IFNA4/IFITM2/IRF2/TRAF3IP2/MLKL/IFNA16/IFNA7/TRIM38/IFNA10/IFNW1/IFNE/TRIM34/SMARCA5/IFNA17/GBP7/IFNL1/IKBKE/CXADR/BST2/MYD88/IFNA6/EXO SC5/TRIM31/CXCL9/APOBEC3F/IFNA14/TBK1/IFNA5/FGL2/IL1B/MORC3/MICB/DUS2/RELA/MOV10/IFNL3/TANK/AZI2/IL12RB1/IFNL2/PTPRC/UNC93B1/RAB2B/NCK1/SIN3A/TSPAN32/CREB3/RIPK3/POLR3C/NLRP1/MAP3K14/APOBEC3H/TRAFF3/BCL2/IL12B/TICAM1/TRIM6/IFNGR2/TRIM65/SEN7/SLFN11/SETD2/RNASE6/PHB2/POLR3H/G3BP1/TLR8/SELENOK/TRIM56/CLPB/TRIM28/POLR3D/AZU1/RPSA/DDIT4/MARCHF2/PHB1/IFNL4/TARBP2/DHX15/ADARB1/UBL7/DHX36/PPM1B/BCL2L1/ATAD3A/ATG16L1/MID2/CARD9/AICDA/BNIP3L/PRF1/ATG5/TRAFF6/POLR3K/RIOK3/STING1/ERCC6/ITCH/HSP90AA1/NDUFAF4/CARD8/RNF26/SPN/EPG5/FOXP3/VAMP8/ZCCHC3/NMB/TREX1/DHX9/BECN1/DNAJC3/APOBEC3C/ITGAX/TRIM13/AKAP1/PYCARD/DDX17/ZDHHC1/AGBL5/UAP1/LYST/RNASEL/PQBP1/LSM14A/EXOC1/POLR3G/ATG7/ZNF175/EXOSC4/DDX1/MICA/SPON2/SERINC5/IRF5/ATG12/MAVS
regulation of viral life cycle	HRV	2.734871	IFIT1/OASL/RSAD2/OAS2/IFIH1/SHFL/ISG15/MX1/OAS1/EIF2AK2/BST2/IFIT5/ZC3HAV1/OAS3/IFITM1/TNF/TRIM5/TRIM25/TRIM22/TRIM21/APOBEC3G/IFNB1/IFI16/PLSCR1/ZNFX1/TRIM6/APOBEC3F/AXL/LY6E/IFITM2/LAMP3/GBP7
	WNV	2.910539	OASL/APOBEC3B/OAS2/ISG20/APOBEC3G/MX1/APOBEC3A/IFITM1/IFIH1/ZC3HAV1/IFIT1/OAS3/CCL5/OAS1/EIF2AK2/PLSCR1/ISG15/IFNA2/TRIM22/TRIM5/ZNFX1/IFITM3/IFI16/IFIT5/APOBEC3D/TNF/TRIM26/RSAD2/LAMP3/SHFL/CXCL8/TRIM25/N4BP1/TRIM21/IFNB1/ADAR/CH25

Gene Ontology Term	Virus	NES	Gene
			H/IFITM2/TRIM38/TRIM34/AXL/GBP7/FMR1/FAM111A/BST2/TRIM31/APOBEC3F/IL32/LY6E/PKN2/ARK2N/TNIP1/NECTIN2/IFNL3/PTX3/LARP1/STAU1/DDX3X/APOBEC3H/BCL2/TRIM6/INPP5K/PPID/CD4/SNX3/TRIM28/BANF1/TOP2A/TARBP2/VPS37B/ADARB1/FCN1/DDX5/LGALS1/RESF1/MID2/AICDA/CD28/TMEM39A/TRIM68/SRPK1/CLEC4G/MPHOSPH8/KPNA2/SMPD1/APOBEC3C/PIIH/TMPRSS2/BSG/RNASEL/SLPI/CD209/MAVS
viral genome replication	HRV	2.645625	IFIT1/OASL/RSAD2/OAS2/IFIH1/SHFL/ISG15/MX1/OAS1/EIF2AK2/BST2/IFIT5/CCL8/ZC3HAV1/OAS3/IFI27/IFITM1/TNF/APOBEC3G/ZC3H12A/IFNB1/IFI16/PLSCR1/ZNFX1/TRIM6/APOBEC3F/IFITM2/GBP7/CCL2
	WNV	2.778649	OASL/APOBEC3B/OAS2/ISG20/APOBEC3G/MX1/APOBEC3A/IFITM1/IFIH1/ZC3HAV1/CCL8/IFI1/OAS3/CCL5/OAS1/EIF2AK2/IFI27/PLSCR1/ISG15/ZNFX1/IFITM3/IFI16/IFIT5/APOBEC3D/CCL2/TNF/RSAD2/ZC3H12A/SHFL/DEK/CXCL8/N4BP1/IFNB1/ADAR/STOM/IFITM2/TRIM38/NFIA/GBP7/ZBED1/FMR1/FAM111A/BST2/APOBEC3F/PKN2/TNIP1/IFNL3/CCNK/LARP1/STAU1/DDX3X/APOBEC3H/BCL2/TRIM6/GRK2/INPP5K/PPID/GAS6/NUCKS1/ATG16L2/TRIM28/BANF1/TOP2A/PHB1/TARBP2/ADARB1/DDX5/ATG16L1/RESF1/AICDA/CD28/ATG5/TMEM39A/FBXL2/HSPA8/SRPK1/LAMTOR5/CTBP1/MPHOSPH8/ZFYVE1/VCP/APOBEC3C/PIIH/RNASEL/MIR221/PCBP1/SMC6/SLPI/CD209/MAVS
antiviral innate immune response	HRV	2.409251	IFIT1/IFIT3/OASL/OAS2/RIGI/IFIT2/IFIH1/MX1/OAS1/EIF2AK2/USP18/CXCL10/OAS3/TNF/TRIM25/DHX58/TRIM6
	WNV	2.442198	IFIT3/RIGI/OASL/OAS2/DHX58/USP18/MX1/ZBP1/IFIH1/IFIT2/IFIT1/OAS3/OAS1/EIF2AK2/TNF/CXCL10/TRIM25/SMARCA5/TRIM31/TBK1/MORC3/DUS2/RELA/NCK1/NLRP1/TRIM6/TRIM65/SEN7/PHB2/CLPB/RPSA/MARCHF2/PHB1/DHX15/ATAD3A/TRA6/STING1/CARD8/NMB/AKAP1/ZDHHC1/UAP1/MAVS
regulation of defense response to virus by host	HRV	2.416820	DTX3L/RIGI/PARP9/PML/STAT1/TRIM22/APOBEC3G/ZC3H12A/CGAS/AIM2/TRIM6/IL15/APOBEC3F/TNFAIP3/LILRB1/IL27
	WNV	2.447801	RIGI/HERC5/APOBEC3G/PARP9/AIM2/CGAS/STAT1/TRIM22/DTX3L/PML/TNFAIP3/IL27/ZC3H12A/IL15/IL23A/LILRB1/NT5C2/TRA63IP2/TRIM38/APOBEC3F/FGL2/IL1B/MICB/IL12RB1/SIN3A/TSPAN32/CREB3/TRA63/IL12B/TRIM6/SELENOK/TARBP2/PPM1B/ATG5/STING1/ERCC6/ITCH/HSP90AA1/RNF26/SPN/FOX3/DHX9/PYCARD/ZDHHC1/PQBP1/ATG12/MAVS
regulation of innate immune response	HRV	2.479576	RIGI/GBP5/IRF1/OASL/DHX58/USP18/GBP2/PARP14/ZBP1/IFIH1/ZC3HAV1/PARP9/AIM2/OAS3/NM/CGAS/NCF1/SERPING1/CCL5/OAS1/EIF2AK2/STAT2/PLSCR1/SERPINB9/ISG15/TRA6D1/IFI35/TRIM5/ZNFX1/IFI16/TNF/LAG3/CALHM6/TNFAIP3/DDX60/CD274/RSAD2/NLRCS/CD300E/DAB2IP/SCIMP/NFKBIZ/TRIM25/ACOD1/LYN/CLEC4E/IRF7/PEL1/N4BP1/EREG/TRIM21/LILRB1/IFNB1/ADAR/CASP1/FPR2/TLR3/NOD2/BIRC3/TLR2/SLC19A1/IRAK2/MEFV/NFKBIA/RIPK2/ZDHHC9/MAP2K6/TNIP3/HSPA1B/HLA-F/RNF125/CYLD/TRIL/HSPA1A/IKBKE/COLEC12/MYD88/TIFAB/TRIM31/KIR2DL4/MNDA/PPARG/RASGRP4/BTK/TASL/FFAR2/IL12A/IRAK3/HCK/TBK1/TICAM2/CASP8/RFTN1/WASHC4/PRKCE/NLRX1/WDFY1/HLA-G/OTUD4/RELA/TNIP1/NECTIN2/PIK3AP1/CD14/UNC93B1/TLR5/RASGRP1/S100A8/SIN3A/TTL12/NOD1/BIRC2/GRAMD4/ADAM8/PTPN2/POLR3C/DDX3X/NLRP1/HLA-B/SRC/HLA-E/IL18RAP/SYK/NAGK/NFE2L2/TRA63/HMGB2/USP15/ERBIN/IL12B/ARRB2/WNT5A/TICAM1/TRIM6/BCL10/ATAT1/SLC15A4/RNF34/TLR1/PSPC1/SH2D1B/CEACAM1/SARM1/AKIRIN2/PHB2/CREBBP/ERAP1/TLR8/TRIM56/CLPB/LACC1/POLR3D/ESR1/TAX1BP1/BANF1/NEK7/SIRT2/APPL1/RBM47/TREM2/PHB1/TARBP2/CLEC6A/IRAK4/APP/NLRP3/FCN1/PTPN1/NR1H3/TNIP2/FGR/NINJ1/ZDHHC18/ZDHHC4/NOP53/LY96/CARD9/TIFA/LYAR/RPS6KA3/PARP1/NPLOC4/CPTP/ATG5/NCR1/TRA66/PTPN22/MATR3/NONO/PUM2/TAB1/ZDHHC5/FBXL2/HSPA8/NLR4/LRRC14/MAPKAPK3/DUSP10/RIOK3/ECSIT/STING1/AURKB/ZNRF1/LAMP2/ITCH/HSP90AA1/CARD8/TLR4/TMIGD3/PPP2CA/EPG5/EP300/SLAMF8/VSIG4/MAPKAPK2/ZDHHC12/ALPK1/CTSS/GFER/USP38/ZCCHC3/HAVCR2/SMPDL3A/CHUK/FYN/UBQLN1/TREX1/DHX9/BECN1/APOE/AP1G1/RAB7B/PJA2/SFPQ/PYCARD/CD300A/RTN4/FAM3A/OGT/CEP63/CSNK1A1/NFKBIL1/S100A14/USP50/PQBP1/PVR/LSM14A/UBE2K/METTL3/PIIM1/YTHDF3/RNF115/CR1/PUM1/MIR146A/POLR3G/GDI1/RPS19/CADM1/STAT5B/SLC46A2/MICA/LRP8/MAPK8/NAGLU/SPSB3/STMP1/TXK/ATG12/MAVS
	WNV	2.577993	RIGI/GBP5/IRF1/OASL/DHX58/USP18/GBP2/PARP14/ZBP1/IFIH1/ZC3HAV1/PARP9/AIM2/OAS3/NM/CGAS/NCF1/SERPING1/CCL5/OAS1/EIF2AK2/STAT2/PLSCR1/SERPINB9/ISG15/TRA6D1/IFI35/TRIM5/ZNFX1/IFI16/TNF/LAG3/CALHM6/TNFAIP3/DDX60/CD274/RSAD2/NLRCS/CD300E/DAB2IP/SCIMP/NFKBIZ/TRIM25/ACOD1/LYN/CLEC4E/IRF7/PEL1/N4BP1/EREG/TRIM21/LILRB1/I

Gene Ontology Term	Virus	NES	Gene
			FNB1/ADAR/CASP1/FPR2/TLR3/NOD2/BIRC3/TLR2/SLC19A1/IRAK2/MEFV/NFKBIA/RIPK2/ZDHHC9/MAP2K6/TNIP3/HSPA1B/HLA-F/RNF125/CYLD/TRIL/HSPA1A/IKBKE/COLEC12/MYD88/TIFAB/TRIM31/KIR2DL4/MNDA/PPARG/RASGRP4/BTK/TASL/FFAR2/IL12A/IRAK3/HCK/TBK1/TICAM2/CASP8/RFTN1/WASHC4/PRKCE/NLRX1/WDFY1/HLA-G/OTUD4/RELA/TNIP1/NECTIN2/PIK3AP1/CD14/UNC93B1/TLR5/RASGRP1/S100A8/SIN3A/TTL12/NOD1/BIRC2/GRAMD4/ADAM8/PTPN2/POLR3C/DDX3X/NLRP1/HLA-B/SRC/HLA-E/IL18RAP/SYK/NAGK/NFE2L2/TRAF3/HMGB2/USP15/ERBIN/IL12B/ARRB2/WNT5A/TICAM1/TRIM6/BCL10/ATAT1/SLC15A4/RNF34/TLR1/PSPC1/SH2D1B/CEACAM1/SARM1/AKIRIN2/PHB2/CREBBP/ERAP1/TLR8/TRIM56/CLPB/LACC1/POLR3D/ESR1/TAX1BP1/BANF1/NEK7/SIRT2/APPL1/RBM47/TREM2/PHB1/TARBP2/CLEC6A/IRAK4/APP/NLRP3/FCN1/PTPN1/NR1H3/TNIP2/FGFR/NINJ1/ZDHHC18/ZDHHC4/NOP53/LY96/CARD9/TIFA/LYAR/RPS6KA3/PARP1/NPLOC4/CPTP/ATG5/NCR1/TRAF6/PTPN22/MATR3/NONO/PUM2/TAB1/ZDHHC5/FBXL2/HSPA8/NLRC4/LRRC14/MAPKAPK3/DUSP10/RIOK3/ECSIT/STING1/AURKB/ZNRF1/LAMP2/ITCH/HSP90AA1/CARD8/TLR4/TMIGD3/PPP2CA/EPG5/EP300/SLAMF8/VSIG4/MAPKAPK2/ZDHHC12/ALPK1/CTSS/GFER/USP38/ZCCHC3/HAVCR2/SMPDL3A/CHUK/FYN/UBQLN1/TREX1/DHX9/BECN1/APOE/AP1G1/RAB7B/PJA2/SFPQ/PYCARD/CD300A/RTN4/FAM3A/OGT/CEP63/CSNK1A1/NFKBIL1/S100A14/USP50/PQBP1/PVR/LSM14A/UBE2K/METTL3/PIIM1/YTHDF3/RNF115/CR1/PUM1/MIR146A/POLR3G/GDI1/RPS19/CADM1/STAT5B/SLC46A2/MICA/LRP8/MAPK8/NAGLU/SPSB3/STMP1/TXK/ATG12/MAVS
innate immune response-activating signaling pathway	HRV	2.326038	RIGI/GBP5/IRF1/OASL/DHX58/GBP2/IFIH1/ZC3HAV1/AIM2/OAS3/NMI/CGAS/OAS1/EIF2AK2/IFI35/IFI16/TNF/TNFAIP3/DDX60/CD274/RSAD2/DAB2IP/SCIMP/NFKBIZ/TRIM25/ACOD1/LYN/CLC4E/IRF7/PELI1/CASP1/TLR3/NOD2/BIRC3/TLR2/SLC19A1/IRAK2/MEFV/NFKBIA/RIPK2/ZDHHC9/MAP2K6/TNIP3/HSPA1B/RNF125/CYLD/TRIL/HSPA1A/COLEC12/MYD88/TIFAB/TRIM31/BTK/TASL/FFAR2/IRAK3/HCK/TBK1/TICAM2/RFTN1/WASHC4/PRKCE/NLRX1/WDFY1/OTUD4/RELA/TNIP1/PIK3AP1/CD14/UNC93B1/TLR5/S100A8/NOD1/BIRC2/GRAMD4/DDX3X/NLRP1/SRC/SYK/NAGK/TRAF3/USP15/ERBIN/ARRB2/TICAM1/BCL10/ATAT1/SLC15A4/RNF34/TLR1/SARM1/PHB2/CREBBP/TLR8/CLPB/LACC1/ESR1/TAX1BP1/BANF1/NEK7/SIRT2/APPL1/TREM2/PHB1/TARBP2/CLEC6A/IRAK4/APP/NLRP3/FCN1/NR1H3/TNIP2/NINJ1/ZDHHC18/NOP53/LY96/TIFA/RPS6KA3/PARP1/NPLOC4/CPTP/TRAF6/PTPN22/PUM2/TAB1/ZDHHC5/FBXL2/HSPA8/NLRC4/LRRC14/MAPKAPK3/RIOK3/ECSIT/STING1/AURKB/ZNRF1/LAMP2/ITCH/CARD8/TLR4/PPP2CA/EPG5/EP300/MAPKAPK2/ZDHHC12/ALPK1/CTSS/ZCCHC3/HAVCR2/SMPDL3A/CHUK/FYN/UBQLN1/TREX1/BECN1/RAB7B/PJA2/PYCARD/CD300A/RTN4/OGT/CSNK1A1/NFKBIL1/S100A14/USP50/LSM14A/RNF115/PUM1/MIR146A/GDI1/SLC46A2/MAPK8/NAGLU/SPSB3/STMP1/MAVS
	WNV	2.490063	RIGI/GBP5/IRF1/OASL/DHX58/GBP2/IFIH1/ZC3HAV1/AIM2/OAS3/NMI/CGAS/OAS1/EIF2AK2/IFI35/IFI16/TNF/TNFAIP3/DDX60/CD274/RSAD2/DAB2IP/SCIMP/NFKBIZ/TRIM25/ACOD1/LYN/CLC4E/IRF7/PELI1/CASP1/TLR3/NOD2/BIRC3/TLR2/SLC19A1/IRAK2/MEFV/NFKBIA/RIPK2/ZDHHC9/MAP2K6/TNIP3/HSPA1B/RNF125/CYLD/TRIL/HSPA1A/COLEC12/MYD88/TIFAB/TRIM31/BTK/TASL/FFAR2/IRAK3/HCK/TBK1/TICAM2/RFTN1/WASHC4/PRKCE/NLRX1/WDFY1/OTUD4/RELA/TNIP1/PIK3AP1/CD14/UNC93B1/TLR5/S100A8/NOD1/BIRC2/GRAMD4/DDX3X/NLRP1/SRC/SYK/NAGK/TRAF3/USP15/ERBIN/ARRB2/TICAM1/BCL10/ATAT1/SLC15A4/RNF34/TLR1/SARM1/PHB2/CREBBP/TLR8/CLPB/LACC1/ESR1/TAX1BP1/BANF1/NEK7/SIRT2/APPL1/TREM2/PHB1/TARBP2/CLEC6A/IRAK4/APP/NLRP3/FCN1/NR1H3/TNIP2/NINJ1/ZDHHC18/NOP53/LY96/TIFA/RPS6KA3/PARP1/NPLOC4/CPTP/TRAF6/PTPN22/PUM2/TAB1/ZDHHC5/FBXL2/HSPA8/NLRC4/LRRC14/MAPKAPK3/RIOK3/ECSIT/STING1/AURKB/ZNRF1/LAMP2/ITCH/CARD8/TLR4/PPP2CA/EPG5/EP300/MAPKAPK2/ZDHHC12/ALPK1/CTSS/ZCCHC3/HAVCR2/SMPDL3A/CHUK/FYN/UBQLN1/TREX1/BECN1/RAB7B/PJA2/PYCARD/CD300A/RTN4/OGT/CSNK1A1/NFKBIL1/S100A14/USP50/LSM14A/RNF115/PUM1/MIR146A/GDI1/SLC46A2/MAPK8/NAGLU/SPSB3/STMP1/MAVS